1. INTRODUCTION

1.1. BACKGROUND OF STUDY

In today's competitive industrial environment, efficient supply chain management plays a crucial role in maintaining production timelines and customer satisfaction. Cesare Bonetti International Pvt.Ltd., a leading manufacturer of high-pressure valves and process equipment, relies on a smooth and well-coordinated supply chain to meet its production demands.

Cesare Bonetti International Pvt.Ltd., Nashik, Maharashtra, is a fully owned Indian subsidiary of Cesare Bonetti S.p.A., Italy, which has more than 100 years of experience in the design & manufacture of high-pressure industrial valves, sight glasses, and level gauges. Cesare Bonetti brand is used extensively in oil & gas, power generation, petrochemical, and chemical process industries, where nothing but the best in terms of reliability, precision, and safety can be accepted.

On the other hand, pre-order management focuses on the proactive processes and systems enabling companies to anticipate, plan for, and coordinate customer orders before formal placement13. This phase involves internal approvals, PR generation, vendor selection, and inter-departmental coordination. The objective of this project is to conduct an in-depth analysis of these crucial phases, from PR to PO, to uncover the root causes of delays and inefficiencies through. Supply chain management plays a vital role in ensuring smooth operations in any manufacturing industry. In particular, the stages before placing an order known as the pre-requisite and pre-ordering phase are crucial in avoiding delays and maintaining efficiency.

Despite its strong global reputation and impressive engineering expertise, Cesare Bonetti International Pvt.Ltd. Faced challenges in the initial stages of its supply chain operations Purchase Requisition (PR) to Purchase Order (PO). Common hurdles like design-related errors, communication breakdowns between the design and plan teams. These chasing departments, delays in securing internal approvals, and a general lack of coordination among teams. These delays not only affect the procurement timeline but also lead to production hold-ups, missed delivery schedules, and increased operational costs. (Krause et al., 2020)

Pre-requisite refers to the essential requirements, conditions, or tasks that must be fulfilled before initiating a particular process, activity, or decision. They serve as foundational steps that ensure smooth and effective progress toward the desired

outcome. In various fields such as education, employment, and project management, pre-requisites play a crucial role in setting the stage for success. For example, in supply chain management, pre-requisites like demand forecasting, inventory assessment, and budget approvals must be completed before raising a purchase order. Without fulfilling these initial requirements, the chances of delays, errors, or inefficiencies increase significantly. Therefore, identifying and addressing pre-requisites is critical for achieving timely and systematic workflow execution.

Pre-order process or set of activities that take place before a formal purchase order (PO) is issued in the supply chain or procurement system. It includes all the planning, approvals, and documentation required to ensure that the actual ordering process runs smoothly and efficiently. In manufacturing and business operations, pre-order activities may involve purchase requisition (PR) preparation, vendor selection, product specification finalization, budget confirmation, and internal approvals. These steps are crucial because they help identify needs, avoid delays, and ensure that the purchase order is accurate and aligned with the organization's requirements. Effectively managing the pre-order phase is essential for timely procurement, cost control, and maintaining uninterrupted production or service delivery. (Alferova, 2024)

This project aims to conduct an in-depth analysis of the pre-requisite and pre- ordering processes within the company’s supply chain function. Through observation, process mapping, and interaction with key departments, the study will identify areas where delays and gaps occur. By adopting an analytical approach, this research will examine how internal systems and team coordination contribute to inefficiencies in the PR to PO cycle.

The pre-requisite phase involves initial tasks such as finalizing design specifications, technical drawings, material requirement planning (MRP), and bill of materials preparation. The pre-ordering phase, on the other hand, includes internal approvals, purchase requisition (PR) generation, vendor selection, and coordination among departments like design, planning, and purchase. Any failure or delay in this critical phase can cause disruptions in procurement, delay production timelines, increase operational costs, and impact on customer satisfaction. (Saragih et al., 2020).

This analytical study undertakes a comprehensive examination of pre-requisites and pre-order management within Cesare Bonetti's supply chain operations1. The pre- requisite phase encompasses foundational elements and initial tasks that must be established before effective supply chain operations can be achieved. These include finalizing

design specifications, technical drawings, material requirement planning (MRP), bill of materials preparation, supplier qualification processes, and the establishment of quality management systems.

The objective of this project is to conduct a comprehensive analysis of the pre- requisite and pre-ordering activities involved in Cesare Bonetti's supply chain management process. By studying the workflow from PR to PO, observing departmental interactions, and identifying operational gaps, the project aims to uncover the root causes of delays. The research takes an analytical approach, mapping the current process, examining internal documentation, and interviewing key stakeholders from relevant departments to gain insights.(Chen et al., 2023)

DURATION -2 Month

DEPARTMENT- Supply Chain Management

ROLL ASSIGNED- PR to PO

1.2.Purpose and importance of the topic-

The purpose of this study is to analyze the pre-requisite and pre-ordering processes within the supply chain of Cesare Bonetti International Pvt. Ltd., focusing specifically on the transition from Purchase Requisition (PR) to Purchase Order (PO). These early-stage procurement activities play a crucial role in determining the efficiency and responsiveness of the overall supply chain. By identifying delays and inefficiencies in these phases, the study aims to develop actionable strategies for process improvement.

The importance of this topic stems from the fact that even minor delays or errors during the PR to PO cycle can cause significant disruptions in manufacturing operations, especially in industries dealing with high-precision products like valves and gauges. Ineffective coordination, poor documentation, and slow internal approvals can lead to production hold-ups, increased operational costs, and missed delivery timelines, all which negatively impacts customer satisfaction and company reputation.(Selvi Ramjan Bee & Bhuvaneswari, 2021)

Understanding and optimizing these early procurement stages is critical for enhancing supply chain agility, cost control, and on-time delivery performance. Moreover, this topic is highly relevant in today's competitive industrial environment, where organizations must streamline internal processes and improve collaboration to ahead.

OBJECTIVE

A. To identify and analyze the key pre-requisite activities (such as demand forecasting, inventory planning, and design finalization) that affect the timely release of purchase orders.

B. To examine the common issues and delays in the pre-ordering phase, particularly in the transition from Purchase Requisition (PR) to Purchase Order (PO).

C. To evaluate the level of coordination and communication between departments such as design, planning, and procurement during the initial stages of the supply chain process.

D. To suggest practical process improvements for reducing delays, improving inter-departmental collaboration, and ensuring smoother execution of procurement operations.

SCOPE OF STUDY

His study is focused on analyzing the pre-requisite and pre-order phases within the supply chain management operations. It specifically covers the internal processes carried out by the design, planning, and procurement departments that contribute to the generation of Purchase Requisitions (PR) and the issuance of Purchase Orders (PO). The scope includes a detailed examination of pre-requisite activities such as demand forecasting, inventory planning, material requirement planning (MRP), bill of materials preparation, and design approvals. Additionally, it encompasses the pre-ordering processes involving internal approvals, vendor selection, and inter-departmental coordination necessary to ensure smooth procurement operations.(Prananta & Hidayat, the study is limited to the company's Nashik facility and focuses on the processes followed over the past 6 to 12 months. It evaluates internal workflows, communication patterns, and departmental collaboration to identify inefficiencies and potential areas for improvement. The research excludes post-order procurement activities such as logistics and supplier delivery, as well as other business functions like finance, customer service, and production operations unless they directly impact the PR to PO cycle. The primary objective is to provide actionable insights that can streamline the company's procurement planning and execution processes.

Method of Data Collection - Observation and Questionnaire ask to employees.

SIGNIFICANCE OF STUDY

The significance of this study lies in its potential to enhance the operational efficiency and strategic effectiveness of supply chain management at Cesare Bonetti International Pvt.Ltd. In industries like high-pressure valve manufacturing, where precision, safety, and timely delivery are non-negotiable, even minor delays in the procurement cycle can lead to serious production setbacks and cost overruns. By focusing on the pre-requisite and pre-order phases specifically the transition from Purchase Requisition (PR) to Purchase Order (PO) this study aims to identify the root causes of inefficiencies that often remain unnoticed in day-to-day operations. The findings will help the company understand how early-stage processes such as demand forecasting, design finalization, inventory planning, and inter-departmental communication affect procurement timelines and overall supply chain performance.

Moreover, the insights generated from this research can support the development of improved workflows, faster decision-making protocols, and stronger coordination among departments. This, in turn, can lead to reduced operational costs, improved supplier relationships, and better adherence to production schedules. The study also provides a foundation for future process automation, training, and policy updates aimed at building a more agile and resilient supply chain. Therefore, this research is not only significant for operational improvement but also for sustaining the company's reputation for quality and reliability in a highly competitive global market.

HYPOTHESIS

A. Hypothesis 1 (Pre-requisite activities)-

a. **H_0-** Demand forecasting, inventory planning, and design finalization do not significantly affect the timely release of purchase orders.

b. **H_1-** Demand forecasting, inventory planning, and design finalization significantly affect the timely release of purchase orders.

B. Hypothesis 2 (PR to PO delay)-

a. **H_0-** Delays in the transition from Purchase Requisition (PR) to Purchase Order (PO) do not impact the overall supply chain efficiency.

b. **H_1-** Delays in the transition from Purchase Requisition (PR) to Purchase Order (PO) negatively impact the overall supply chain efficiency.

C. Hypothesis 3 (Departmental coordination)-

a. **H_0-** Coordination between design, planning, and procurement ordering process.

b. **H_1-** Better coordination between design, planning, and procurement departments improves the speed of the pre-ordering process.

D. Hypothesis 4 (Pre-ordering and delivery performance)-

a. **H_0-** Improving the pre-ordering process does not lead to better on-time delivery of products.

b. **H_1-** Improving the pre-ordering process leads to better on-time delivery of products.

1.5 Limitations of the Study

While this study provides valuable insights into the pre-requisite and pre-ordering processes of the supply chain at Cesare Bonetti International Pvt. Ltd., several limitations must be acknowledged:

A. Restricted Data Access: Due to confidentiality and company policy, access to certain procurement data, internal reports, and real-time performance metrics was limited. This may have affected the depth of some analyses.

B. Time Constraint: The research was conducted over a limited duration of two months, which constrained the ability to observe long-term trends or seasonal variations in procurement cycles.

C. Limited Sample Size: Data collection was based on interactions with a select number of employees from key departments. The insights, while useful, may not fully represent the experiences and challenges of all stakeholders involved.

D. Reliance on Self-Reported Information: Some of the data collected through questionnaires and interviews may be subject to bias, as responses reflect individual perceptions and may not always be fully objective.

E. Dynamic Work Environment: Ongoing improvements or changes in procurement tools and systems during the study period may have influenced the consistency of observations and findings.

F. Scope Limitation: The study focuses only on internal processes related to PR to PO within the Nashik facility. It excludes post-order activities such as logistics, vendor performance, and supply chain stages outside of procurement.

THEORETICAL BACKGROUND OF THE STUDY

In today's fast-paced and highly competitive industrial landscape, **Supply Chain Management (SCM)** plays a crucial role in shaping how organizations perform, manage costs, and keep their customers satisfied. This study looks specifically at the early stages of the procurement cycle before and during the ordering process and explores how these steps impact the overall supply chain.(Fitriyani et al., 2022)

At the heart of the analysis are theories from **process management, operations, and organizational coordination**. These provide a foundation for understanding how businesses can better manage the transition from **Purchase Requisition (PR)** to **Purchase Order (PO), a point** in the process that often becomes a bottleneck. If this transition isn't handled smoothly, it can cause costly delays and disrupt production.

The **pre-requisite phase** involves key planning activities like demand forecasting, inventory checks, and material requirements planning (MRP). It also includes preparing necessary technical documents, such as design specifications and bills of materials. This part of the process leans heavily on classical operations research, where aligning available resources with planning outputs is essential for staying on schedule.

After that comes the **pre-ordering phase**, which includes things like internal approvals, evaluating vendors, and communicating between departments. Organizational theory, especially Galbraith's Information Processing Theory suggests that when teams like design, procurement, and planning aren't aligned, it can cause breakdowns in communication and coordination. Supporting this view, Alferova (2024) stresses the importance of having a clear and traceable pre-order management system to reduce delays and improve performance across the supply chain.

Finally, the study draws from **lean supply chain principles**, which focus on cutting waste, standardizing processes, and removing steps that don't add value. Many of the delays seen in this research such as missing drawings, documentation errors, and long approval times show just how relevant these ideas are in real-world manufacturing settings

1.6 LITERATURE REVIEW

Imran Munir, Kashif Mahmood, Zaeem Shafqat, Muniba Ghaffar Procurement 4.0 and Sustainable Supply Chain Performance: The Mediating Role of

Procurement Process Optimization (PPO) (2023): This study explores the complex relationships between Procurement 4.0, sustainable supply chain performance, and procurement process optimization in modern supply chain management. Focusing on food manufacturers in Pakistan, it uses PLS-SEM to test hypotheses and the theoretical framework. The research highlights how procurement process optimization mediates the link between Procurement 4.0 strategies and improved sustainability outcomes in the supply chain. Additionally, a business process simulation demonstrates how automation in Procurement 4.0 enhances the procurement cycle and sustainability performance. The study provides a strategic foundation for future research in sustainable supply chain management (SSCM) and sustainable supply chain performance (SSCP), addressing current trends and gaps.

Naomy Kemunto Okinyi, Dr. Evans Kwendo, Dr. Kiongera Fredrick N, Effect of Pre-Contracting Procurement Practices on Supply Chain Performance of Road Authorities in Kakamega County, Kenya (2023): This study examined the impact of pre-contracting procurement practices on the supply chain performance of road authorities in Kakamega County, Kenya, including KeNHA, KURA, and KeRRA. Despite challenges in adopting procurement practices due to complexity and resource constraints, the research aimed to understand how pre-contracting influences supply chain outcomes. Using system theory and causal research methods, data from 48 supply chain, road, and account officers were collected via structured questionnaires and analyzed with descriptive and inferential statistics. Results from regression analysis showed that pre-contracting procurement practices had no significant effect on supply chain performance ($t = 0.605$, $P > 0.05$), contributing only 0.8% to performance changes. The study suggests that these institutions should review and improve their procurement practices to better enhance supply chain performance.

Aisha Ahmed Madugu, Babangida Bature , Ibrahim M. Idris, and Mustapha Lawal Malumfashi,A model for Optimal Pricing and Ordering Strategies for Perishable Goods with Delayed Deterioration, Two-Stage Demand, and Partial Backorders under Delayed Payment Acceptance(2024): This study presents an Economic Order Quantity (EOQ) model for items that deteriorate over time and have two-phase demand rates, linear holding costs, partial backlogging, and two-level pricing under a trade credit policy. Demand is time-dependent and quadratic before deterioration, then constant afterward. The model allows for shortages and aims to

determine the optimal inventory time, cycle length, and order quantity to maximize total profit. Analytical solutions, numerical examples, and sensitivity analyses are provided. Results show that higher deterioration rates lead to lower optimal inventory time, cycle length, order quantity, and profit. Therefore, minimizing deterioration is crucial for maximizing profitability.

Julia Krause, Andreas Kaufmann, Dirk Riehle, The Code System of a Systematic Literature Review on Pre-Requirements Specification Traceability (2020): The study analyses the effectiveness of a specialized e-commerce system designed to handle pre-order fashion sales in Indonesia, focusing on Sakinah_id SMEs. Fashion products dominate the online market but lack efficient pre-order processing systems. Traditional methods lead to delays and prolonged customer interactions. The implemented e-commerce solution significantly improved the pre-sales process by speeding up customer service and enabling immediate sales quotations. Both buyers and SME employees reported time savings, especially during high order volumes, reducing the workload and increasing overall efficiency.

Jaswadi, Andi Kusuma Indrawan, Reteaming Ayu Dyah Sawitri Implementation of Pre-Order Via Request Item Sales Using Carefully Designed Pre-Sales Steps to Enhance E-Commerce Effectiveness: An Indonesian SME Case Study (2020): The study evaluates a specialized e-commerce system developed for handling pre-order fashion sales in Indonesia, focusing on Sakinah_id SMEs. While fashion products are highly popular online, traditional systems fail to support customized pre-orders efficiently, causing delays in the pre-sales process. The implemented e-commerce platform significantly improved this stage by speeding up customer interactions and enabling immediate sales quotations. Both buyers and SME employees reported shorter processing times, especially during high order volumes, reducing employee workload and improving overall efficiency.

Ana Cristina Ferreira Ângela Silva, Supplier selection and procurement in SMEs: insights from the literature on key criteria and purchasing strategies (2022): This study explores how strategic purchasing and supplier selection can enhance the supply processes of SMEs. It identifies key purchasing strategies such as cost reduction, risk management, global sourcing, total quality management, sustainable management, and supplier management. Through a literature review and content analysis of 59 Scopus-indexed documents, the research highlights the top six

supplier selection criteria: quality of goods, delivery time compliance, price/cost, supplier reputation, geographical location, and supplier performance history. These criteria support long-term company goals and should be integrated into strategic planning. The study emphasizes that simple, cost-effective strategic purchasing practices can significantly improve SME procurement by reducing lead times and logistics costs, especially in the context of environmental concerns, technological advancements, and global sourcing risks.

Rosita Fitriyani, Yuli Liestyana, Agung Satmoko, an analysis of supply chain management practices in MSMEs (2022): This study examines how supply chain management (SCM) practices affect the performance of micro, small, and medium enterprises (MSMEs). Using interviews and questionnaires from 50 MSMEs that have adopted SCM, the research analyzes key SCM aspects: customer relationship management, supplier relationship management, goal congruence, and information sharing. Results show that these SCM practices positively impact both firm and supply chain performance. However, supply chain performance does not mediate the relationship between SCM practices and overall firm performance. Notably, supplier relationship management scored the lowest, indicating a need for MSMEs to strengthen collaboration with suppliers by improving raw material quality and continuously enhancing products.

Alif Abi Hanif, Rakhmat Ceha, Chaznin R. Muhammad , Perbaikan Kinerja Pengadaan Material Perusahaan dengan Mengunakan Pendekatan Supply Chain Operation Reference (SCOR) dan Lean Supply Chain (2024): This study evaluates the material procurement performance of PT IMP, a metal frame manufacturer in Cikarang, using the Supply Chain Operation Reference (SCOR) model, which includes five core processes: plan, source, make, deliver, and return. The current performance, based on company monitoring systems, falls within the "average" range (score 50–70). To enhance this performance, the study proposes strategic improvements focused on 25 key performance indicators (KPIs) that influence material procurement effectiveness.

Selvi. J. Ramjan Bee, Dr. K. Bhuvaneswari, Impact of Sustainable Procurement Practices Inducing Supply Chain Management's Stewardship (2024): This study explores the integration of sustainable procurement strategies into supply chain management, focusing on their impact on social responsibility, economic efficiency,

and environmental performance. Key strategies analyzed include life cycle assessment, green purchasing, and supplier evaluation. Through a detailed literature review, the study identifies both the benefits and challenges of these approaches. Findings show that while sustainable procurement can significantly improve sustainability outcomes, successful implementation requires a comprehensive and well-structured approach.

Yoshiatsu Kawabata, Yuta Hosokawa, and Katsuhide Fujita, Detection of Bottleneck in Manufacturing Supply Chain using Specific KP (2022): This study explores how digital transformation (DX) enables the collection and analysis of real-time data across the entire supply chain (SC) network to improve management and identify bottlenecks (BNs). Using simulation tools that model SC operations and track key performance indicators (KPIs) like average lead time and its standard deviation, bottlenecks can be detected regardless of production methods (push or pull). BNs are identified through work-in-process (WIP) buildup during high demand and node utilization during low demand. Additionally, the study introduces a method to rank how close each node is to becoming a bottleneck based on utilization rates.

2. ORGANIZATION PROFILE

2.1.INDUSTRY OVERVIEW

The manufacturing industry is a cornerstone of economic development, encompassing the production of tangible goods through various processes. It spans sectors such as automotive, electronics, textiles, food and beverages, and more. Manufacturing involves raw material acquisition, processing, assembly, and distribution. Automation and technology have revolutionized the sector, leading to increased efficiency and precision. However, challenges like supply chain disruptions and environmental concerns have emerged. Advanced techniques like additive manufacturing (3D printing) have gained prominence. The industry contributes significantly to GDP and employment, driving innovation and fostering economic growth. Global competition drives companies to constantly innovate, improve quality, and reduce costs. The shift towards sustainable practices and smart manufacturing is gaining traction, emphasizing resource efficiency and reduced waste. In recent years, industry has experienced transformation due to the rise of e-commerce and changing consumer preferences. As technologies like AI, IoT, and robotics advance, the future of manufacturing is likely to be more connected, data-driven, and adaptable.

fig.2.1

The manufacturing industry is the sector of the economy that produces goods through the transformation of raw materials, parts, and components. It is a broad sector that includes a wide range of activities, from the production of basic materials such as steel and cement to the assembly of complex products such as automobiles and aircraft.

The manufacturing industry plays a vital role in shaping the global economy and supporting livelihoods across the world. In 2020, it contributed approximately 16.9% to the world's GDP, reflecting its immense economic influence. Beyond numbers, manufacturing is the backbone of employment for millions offering not just jobs, but stability, skill development, and opportunities for communities in both developed and emerging economies. The manufacturing industry is constantly evolving, driven by technological innovation and changing consumer demands. In recent years, the industry has been undergoing a digital transformation, with the adoption of new technologies such as robotics, artificial intelligence, and big data. This has led to increased productivity and efficiency in manufacturing, as well as the creation of new products and services. The manufacturing industry is facing several challenges, including rising labor costs, increasing competition from low-cost countries, and the need to meet environmental regulations. However, industry is also seeing opportunities, such as the growth of e-commerce and the demand for customized products. (Cesare Bonetti, 2024)

2.1.1.MAJOR PLAYER- (Cesare Bonetti, n.d.)

A. L&T Valves Limited (Chennai)
B. Forbes Marshall (Pune)
C. KSB Limited (Pune & Coimbatore)
D. BHEL (Bharat Heavy Electricals Limited) – Valves Division (Tiruchirappalli)
E. Hyper Valves Pvt.Ltd. (Ahmedabad)
F. Amtech Valves (Ahmedabad)
G. Sap Industries Ltd. (Ahmedabad)

2.2. COMPANY PROFILE

fig.2.2

COMPANY NAME- Cesare Bonetti International Pvt Ltd.

CESARE BONETTI
QUALITY VALVES & LEVEL GAUGES

DESCRIPTION- The company is a manufacturer of industrial valve products such as instrumentation valves, fittings, forging and thermowells, needle valves, and pressure gauge valves.

PRODUCTS & SERVICES- Instrumentation valves, fittings, forging and Thermowells.

CURRENT STATUS- The status of Cesare Bonetti International Private Limited is – **Active.**

AGM- The last reported AGM (Annual General Meeting) of Cesare Bonetti International Private Limited, was held on 10 November 2025.

CORPORATE IDENTIFICATION NUMBER (CIN)-

U27320MH2017PTC299070

INCORPORATION DATE -AGE: 28 August 2017 / 9 Years.

TYPE- Unlisted Private Company

CATEGORY- Company limited by Shares

SUBCATEGORY- non-govt company

Email- cbindia@c-bonetti.com

Website- https://c-bonetti.com/

ADDRESSES-

FACTORY-

Cesare Bonetti International Pvt. Ltd.

Gat No. 135 To 140 & 142 To 148 At Village Gonde Dumale, Taluka Igatpuri, Dist. Nashik – 422403.

REGISTERED OFFICE-

Cesare Bonetti International **Pvt. Ltd.** ASA-19, Ashirwad Bangalow, Ashwin Nagar, Nashik – 422009 Maharashtra, India

fig 2.3

2.3. MISSION OF CBIPL

With dedication and consistency, we strive to support our customers in every situation delivering quality, safety, and efficiency that stand the test of time. Since our founding, we've been designing and manufacturing valves and level gauges for industrial use, while continually investing in our R&D, facilities, and customer service to better serve the market.

Our mission is simple: to stay as close as possible to both long-standing and new customers, listen closely to their needs, share their challenges, and work together to find the best possible solutions. We aim not only to meet expectations but to exceed them by anticipating needs, offering genuine support, and building strong, lasting relationships grounded in trust and collaboration for the future. ((Cesare Bonetti, 2025))

2.4. VISION CBIPL

Our long-standing experience is one of the cornerstones of our success. Tradition, expertise, and innovation are woven into our daily work, they define who we are. We are proud of our history and the invaluable knowledge it holds, and we are committed to preserving this legacy, expanding it, and passing on its core values to future generations. ((Cesare Bonetti, 2025))

Our journey has been enriched by a constant exchange of ideas, learning, and collaboration. Every product we create reflects our alignment with market needs and the evolving landscape of technical advancement. For us, relationships with our customers, colleagues, and suppliers are not just about the present; they are built with the future in mind. We aim to grow together by fostering strong, lasting partnerships rooted in mutual respect, environmental responsibility, safety, and shared values.

2.5. A LEADING VALVE MANUFACTURER IN THE WORLD

Cesare Bonetti International Pvt.Ltd. is a globally recognized manufacturer of high-performance valves and liquid level gauges and indicators. We specialize in both standard and custom-engineered solutions tailored to the specific requirements of any fluid pressure and temperature handling application. Our reputation for quality, reliability, and exceptional service is known and trusted worldwide. ((Cesare Bonetti, n.d.))

2.6. QUALITY VALVES & LEVEL GAUGES

We offer the largest range (from ¼" NB up to 48″ NB and up to ASME Class 4500) of High-Pressure Globe, Gate and Check Valves, Bellows Sealed Valves, High Pressure-High Temperature and/or Cryogenic service Ball Valves, world renewed and recently improved Piston Valves and the widest pressure, temperature, and application range of Glass, Magnetic, and Resistive Probe Liquid Level Gauges for every possible liquid handling application.

MAJOR CLIENTS

A. Technip
B. Statoil
C. Siemens
D. Shell
E. Saipem

2.7 COMPANY CERTIFICATE

fig 2.4

TÜV NORD

CERTIFICATE

Management system as per
ISO 9001 : 2015

CESARE BONETTI INTERNATIONAL PRIVATE LTD.
Gut No: 135 to 140 & 142 to 149, Village Gonde Dumala,
Taluka : Igatpuri, Dist : Nashik – 422 403,
Maharashtra,
India

Design, Development, Manufacture & Supply of Industrial Valves, Manifolds and Level Gauges

Certificate Registration No.
Audit Report No.
Valid until

Certification Body
at TÜV NORD CERT GmbH

Place : Mumbai

This certification was conducted in accordance with the TÜV NORD CERT auditing and certification procedures.
It shall be valid subject to regular Surveillance Audits.

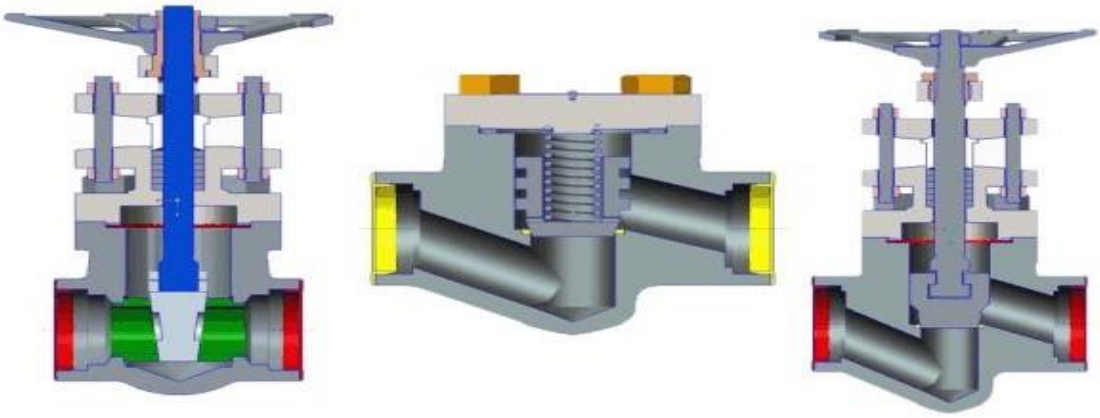
For TÜV NORD CERT GmbH
Langenscheidtstraße 20
45144 Essen
www.tuv-nord-cert.com

TÜV India Pvt. Ltd., 301, Ganga Plaza - I, J.R.D. Marg, Chembur (W), Mumbai - 400 089, India www.tuv-nord.com

IAF DAkkS

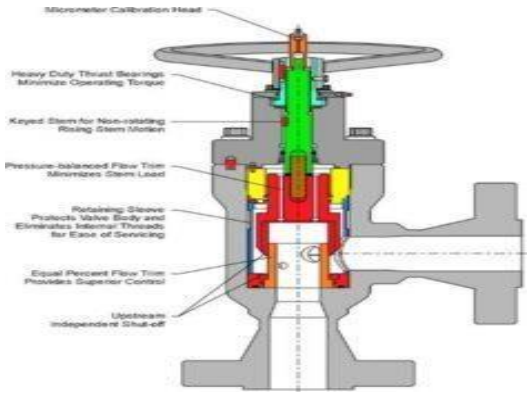
2.8. PRODUCTS OF CBIPL

Ball valve fig 2.5

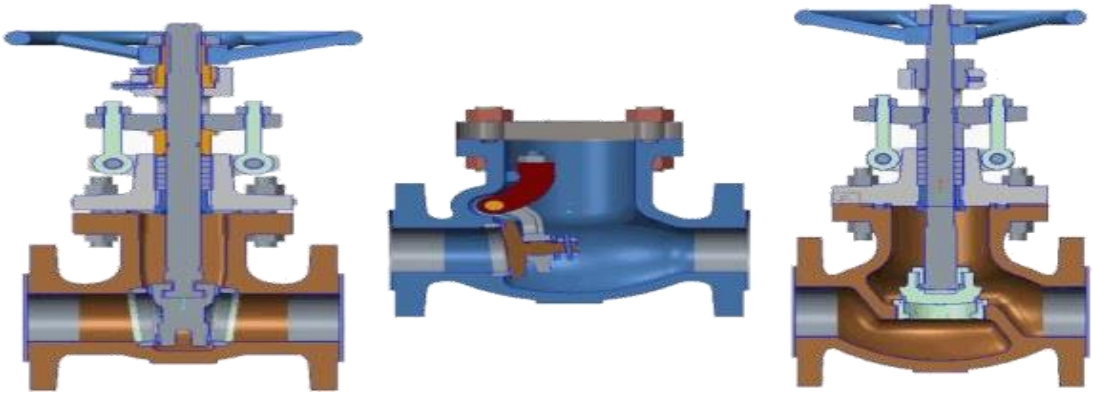


Gate valve Table fig2.6

Globe valve Table fig2.7



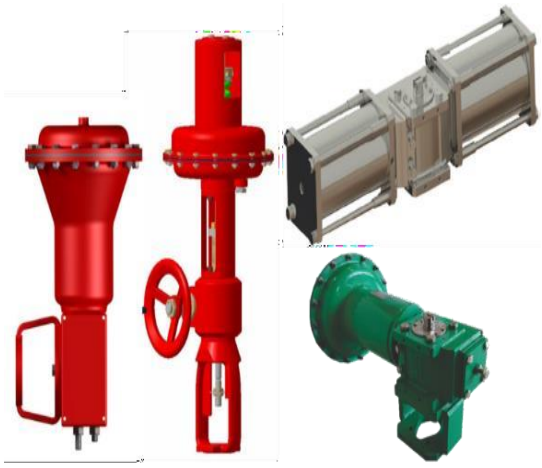
Gate Valve fig 2.8



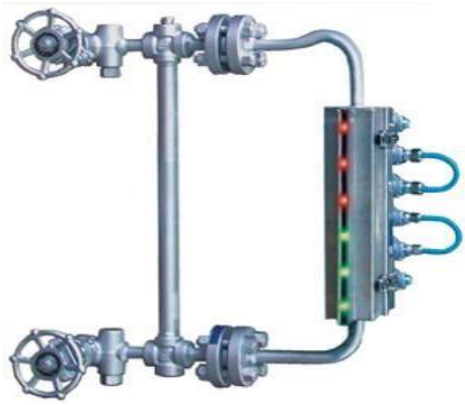
Gate Valve fig 2.9

Christmas Tree fig 2.10

Manifold System fig 2.11



Blowout Preventer fig 2.12



Butterfly Valve fig 2.13

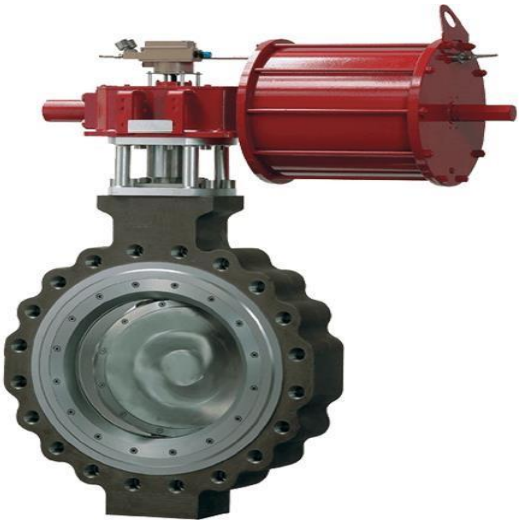
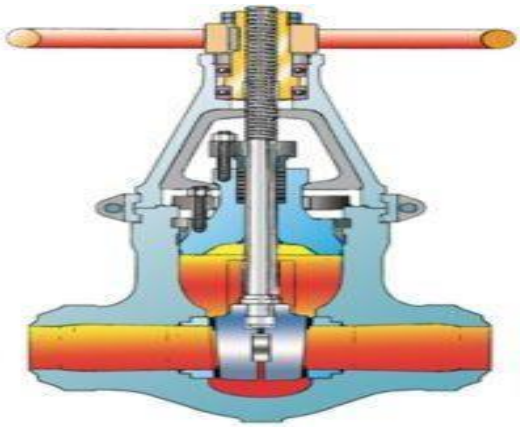


fig2.14



2.9. ORGANIZATION CHART

fig 2.15

3. PROBLEM TASK OUTLINE

3.1. PROBLEM STATEMENT

In the highly specialized and quality-sensitive manufacturing sector, effective supply chain management (SCM) is critical to ensuring timely production and delivery.at Cesare Bonetti International, which deals with precision industrial components such as high-pressure valves, even minor disruptions or inefficiencies in the supply chain can lead to production delays, increased costs, and customer dissatisfaction.

Despite having a structured SCM system in place, the company faces challenges related to the identification and implementation of essential pre-requisites such as accurate demand forecasting, inventory control, supplier coordination, and procurement planning. Furthermore, the pre-ordering process critical for ensuring timely availability of raw materials and components may suffer from inefficiencies such as misalignment with production schedules, communication gaps with suppliers, or delays in approvals.

3.2. OBJECTIVES OF THE ASSIGNED WORK

A. To identify and analyse the key pre-requisite activities—such as demand forecasting, inventory planning, and design finalization—that influence timely procurement.

B. To evaluate common causes of delay in the pre-order process, particularly during the PR to PO transition.

C. To assess the level of coordination and communication between departments such as design, planning, and procurement.

D. To suggest practical improvements for optimizing early-stage procurement processes and enhancing overall supply chain responsiveness.

3.3. SCOPE OF THE TASK/PROJECT

A. The pre-requisite phase: including MRP (Material Requirement Planning), finalization of specifications, and bill of materials preparation.

B. The pre-order phase: covering PR generation, internal approvals, vendor selection, and PO issuance.

C. Process mapping and stakeholder interviews within departments such planning.

D. The project excludes downstream supply chain activities such as logistics, vendor delivery, and post-order tracking.

3.4 IMPORTANCE AND RELEVANCE TO THE ORGANIZATION

In a company like Cesare Bonetti, which operates in high-specification, time-sensitive industries such as oil & gas and power, procurement efficiency is directly linked to business success. Identifying and addressing inefficiencies in the PR-to-PO process is vital to reducing production downtime, improving cost control, and ensuring customer commitments are met. This project supports the company's goals of achieving supply chain excellence and enhancing internal operational coordination.

3.5 CONSTRAINTS, CHALLENGES, OR LIMITATIONS

A. **Limited access to live procurement data** due to confidentiality and approval restrictions.

B. **Time constraints** in completing a thorough end-to-end analysis within the project duration.

C. **Dependence on availability of staff** for interviews and process mapping across departments.(Okinyi et al., 2023)

D. **Potential bias** in self-reported data or departmental interpretations of issues.

E. **Dynamic work environment**, where parallel improvements or changes in procurement tools may affect the consistency of findings.

4. RESEARCH METHODOLOGY

4.1. RESEARCH & ITS TYPE

This study is both **analytical** and **descriptive** in nature. It focuses on uncovering the root causes of delays in the **pre-requisite** and **pre-ordering** stages of the supply chain at Cesare Bonetti International Pvt. Ltd. The aim is to closely examine internal workflows, inter-departmental coordination, and overall process efficiency. To ensure a well-rounded understanding, the research adopts a **mixed-method approach** using quantitative tools like frequency analysis and chi-square tests, alongside qualitative methods such as interviews and field observations.

4.2. STUDY DURATION

The study was conducted over a **two-month period**,(15 May 2025-15 July 2025)during which key departments involved in the PR-to-PO (Purchase Requisition to Purchase Order) process were closely observed, surveyed, and analyzed.

4.3. OBJECTIVES

The main objectives of this study are:

A. To identify and analyze critical **pre-requisite activities** such as demand forecasting, inventory planning, and design finalization that influence timely procurement.

B. To examine common **delays** during the PR to PO transition.

C. To evaluate the quality of **communication and coordination** between departments like Design, Planning, and Procurement.

D. To provide **recommendations** for improving process efficiency and inter-departmental responsiveness.

4.4. RESEARCH DESIGN FRAMEWORK

A structured research framework was adopted to maintain clarity, coherence, and depth throughout the study. The process involved the following steps:

A. **Problem Identification**

Recognizing inefficiencies and delays in the PR to PO workflow.

B. **Literature Review**

Reviewing current industry practices and scholarly work related to lean supply chains, digital procurement, and organizational coordination.

C. **Objective & Hypothesis Formation**

Developing clear objectives and testable hypotheses about potential process bottlenecks and coordination issues.

D. **Data Collection**

a. **Primary Data**: Collected via surveys, direct field observations, and interviews with key stakeholders.

b. **Secondary Data**: Sourced from company records like PR/PO logs, departmental reports, and internal audit logs.

E. **Data Analysis**

a. Descriptive statistics to understand patterns

b. Chi-square tests to find relationships between variables

c. ANOVA to compare mean delays across departments or delay types

F. **Findings Interpretation**

Interpreting results to validate or refute the stated hypotheses.

G. **Strategic Recommendations**

Suggesting practical improvements such as ERP-based process automation, updated Standard Operating Procedures (SOPs), and training initiatives.

4.5. SAMPLING STRATEGY

A. To ensure meaningful insights, the study adopted a **purposive sampling** approach targeting individuals who are directly involved in the supply chain processes.

B. **Sampling Method**: Non-probability, purposive sampling

C. **Sample Size**: 32 participants across five departments—Design, Planning, Procurement, Purchase, and Inventory

D. **Participant Profile**: A diverse mix of junior officers to senior managers, including Engineers, Assistant Managers, and Department Heads

E. **Data Collection Tools**

a. Structured questionnaires focused on supply chain activities

b. Observational field notes

c. Targeted interviews with department-specific personnel

F. Tools Use

a. Jamovi (to test run like frequencies table chi-square, ONE-WAY ANOVA, frequencies)

b. Excel (to use pivot chart, for graph)

Data Sources

A. **Primary Data**: Derived from field interactions providing direct, real-time insights into process workflows.

B. **Secondary Data**: Sourced from internal documents like PR/PO logs, departmental workflows, and offering valuable context and supporting evidence.

74.6. PRIMARY DATA

Descriptive

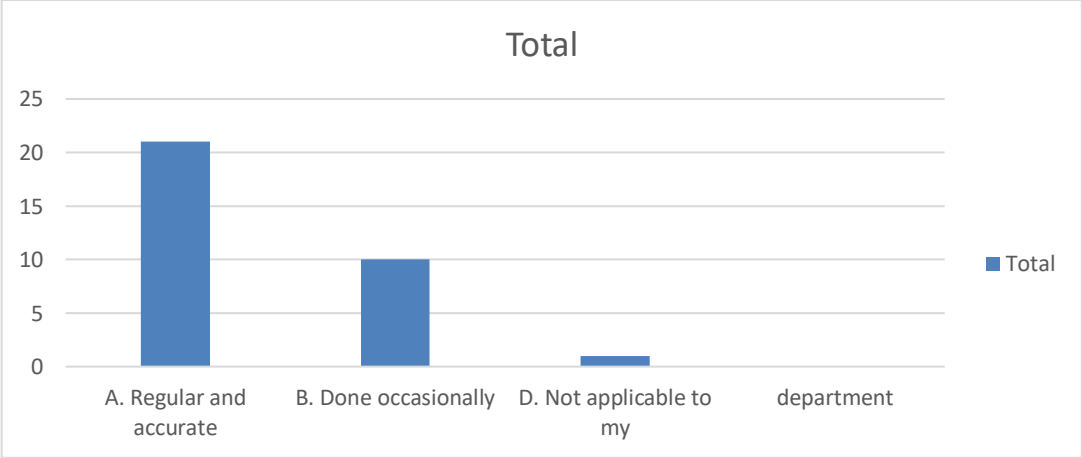
Table 4.6.1

Frequencies of 1. What is the status of demand forecasting in your department?

Row Labels	Counts	% of Total	Cumulative %
A. Regular and accurate	21	65.6%	65.6%
B. Done occasionally	10	31.3%	96.9%
D. Not applicable to my department	1	3.1%	100.0%

Graph 4.6.1

Forecasting



Interpretation:

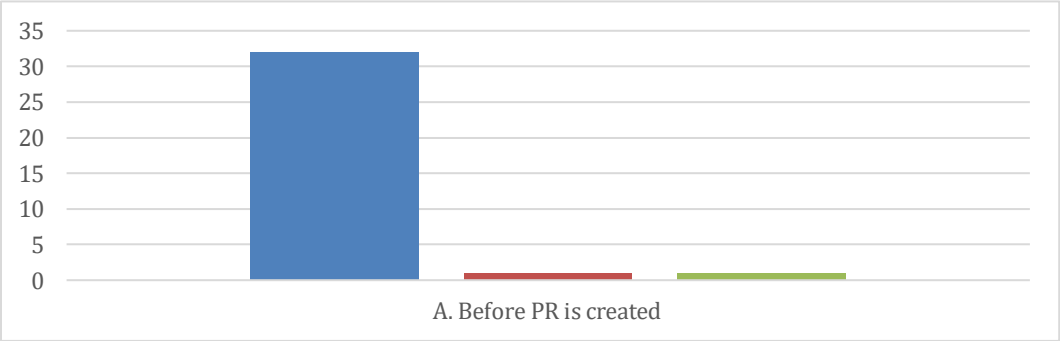
Most respondents (65.6%) reported that demand forecasting is conducted regularly and accurately in their department, indicating a strong emphasis on planning. However, 31.3% stated it is done only occasionally, and 3.1% said it's not applicable, suggesting some inconsistency across departments.

Table 4.6.2
Frequencies of 2. When is inventory planning usually done?

Row Labels	Counts	% of Total	Cumulative %
A. Before PR is created	32	100.0%	100.0%

Graph 4.6.2

Inventory Planning



Interpretation:

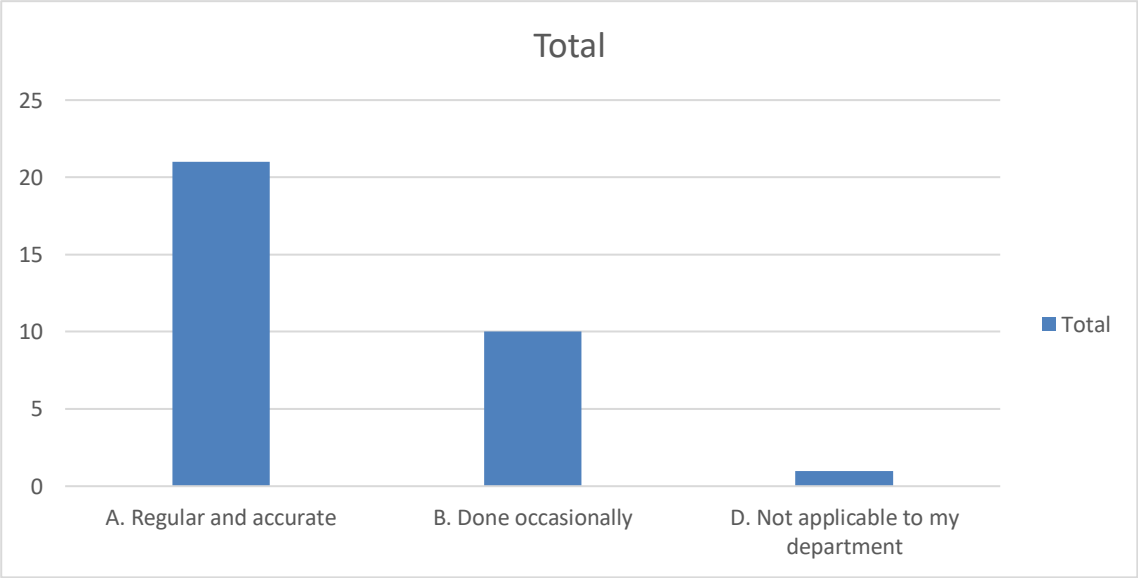
All respondents (100%) indicated that inventory planning is done before the Purchase Requisition (PR) is created. This shows a standardized and proactive approach to inventory management across all departments.

Table 4.6.3
Frequencies of 3. When is design finalization usually completed?

Row Labels	Counts	% of Total	Cumulative %
A. Regular and accurate	21	65.6%	65.6%
B. Done occasionally	10	31.3%	96.9%
D. Not applicable to my department	1	3.1%	100.0%

Graph 4.6.3

Design Finalization



Interpretation:

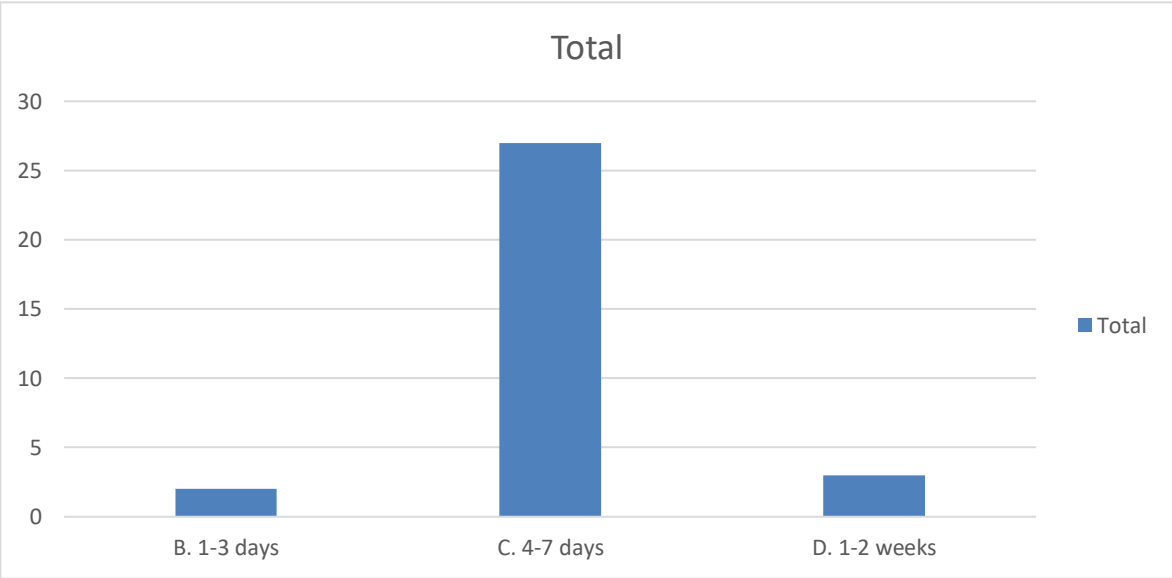
The majority (65.6%) reported that design finalization is completed regularly and accurately, indicating a well-structured design process in most departments. However, 31.3% noted it is done only occasionally, and 3.1% found it not applicable, highlighting variability in design-related responsibilities.

Table 4.6.4
Frequencies of 4. What is the average processing time from PR creation to PO release

Row Labels	Counts	% of Total	Cumulative %
B. 1-3 days	2	6.3%	6.3%
C. 4-7 days	27	84.4%	90.6%
D. 1-2 weeks	3	9.4%	100.0%

Graph 4.6.4

Processing Time



Interpretation:

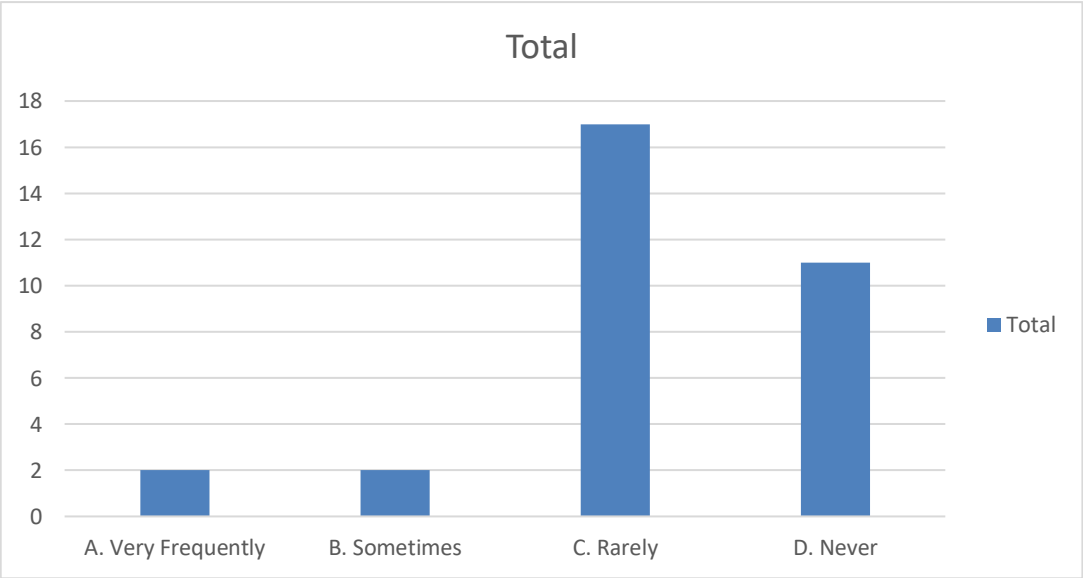
A large majority (84.4%) of respondents reported that the processing time from PR creation to PO release takes 4–7 days, indicating an efficient procurement cycle. Only a small portion experiences shorter (6.3%) or longer (9.4%) durations, suggesting overall consistency with minor delays in some cases.

Table 4.6.6
Frequencies of 6. How often do PRs get delayed in approval?

Row Labels	Counts	% of Total	Cumulative %
A. Very Frequently	2	6.3%	6.3%
B. Sometimes	2	6.3%	12.5%
C. Rarely	17	53.1%	65.6%
D. Never	11	34.4%	100.0%

Graph 4.6.6

Delayed



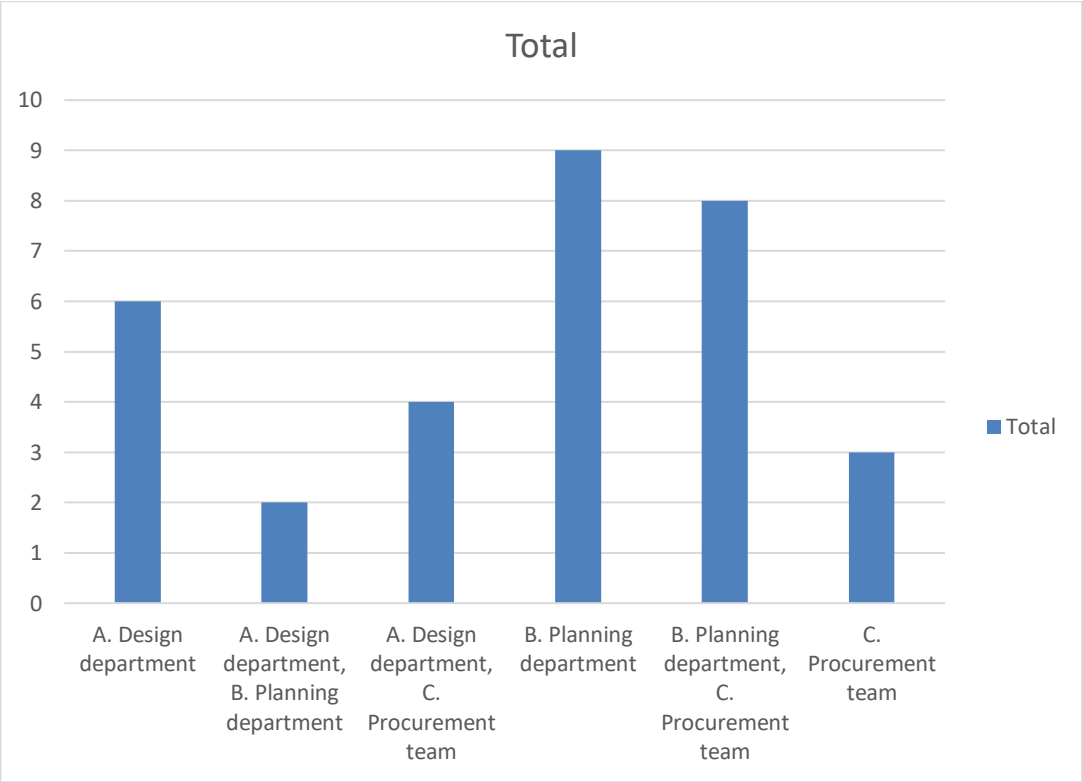
Interpretation:

Most respondents (53.1%) indicated that PR approvals are rarely delayed, and 34.4% reported they are never delayed, reflecting a generally smooth approval process. Only 12.6% experienced frequent or occasional delays, suggesting minimal disruption in most departments.

Table 4.6.8
Frequencies of 8. Who typically initiates the follow-up for delayed approvals

Row Labels	Counts	% of Total	Cumulative %
A. Design department	6	18.8%	18.8%
A. Design department, B. Planning department	2	6.3%	25.0%
A. Design department, C. Procurement team	4	12.5%	37.5%
B. Planning department	9	28.1%	65.6%
B. Planning department, C. Procurement team	8	25.0%	90.6%
C. Procurement team	3	9.4%	100.0%

Graph 4.6.8
Delayed Approvals



Interpretation:

The follow-up for delayed approvals is most initiated by the Planning department (28.1%) either alone or in coordination with others. A significant share also involves joint efforts, especially with the Procurement team (25%). This suggests that follow-up responsibilities are shared across departments, with Planning playing a central role.

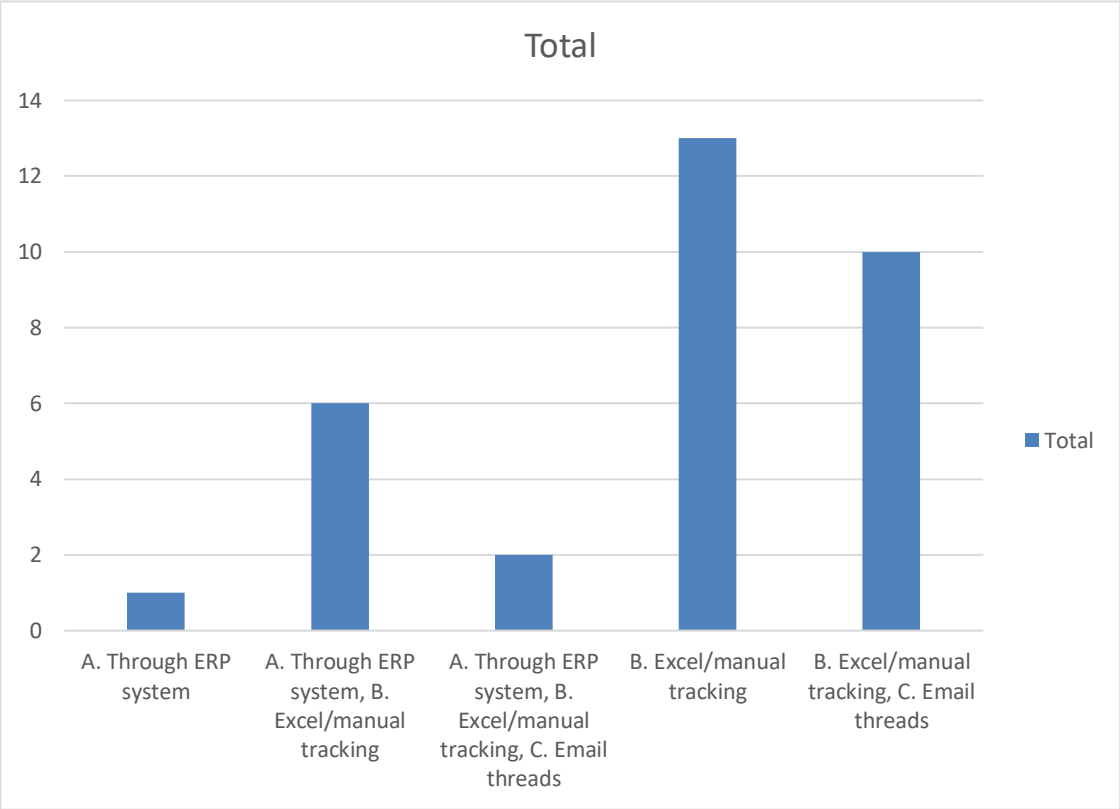
Table 4.6.9

Frequencies of 9. How is the PR to PO tracking done?

Row Labels	Counts	% of Total	Cumulative %
A. Through ERP system	1	3.1%	3.1%
A. Through ERP system, B. Excel/manual tracking	6	18.8%	21.9%
A. Through ERP system, B. Excel/manual tracking, C. Email threads	2	6.3%	28.1%
B. Excel/manual tracking	13	40.6%	68.8%
B. Excel/manual tracking, C. Email threads	10	31.3%	100.0%

Graph 4.6.9

Tracking



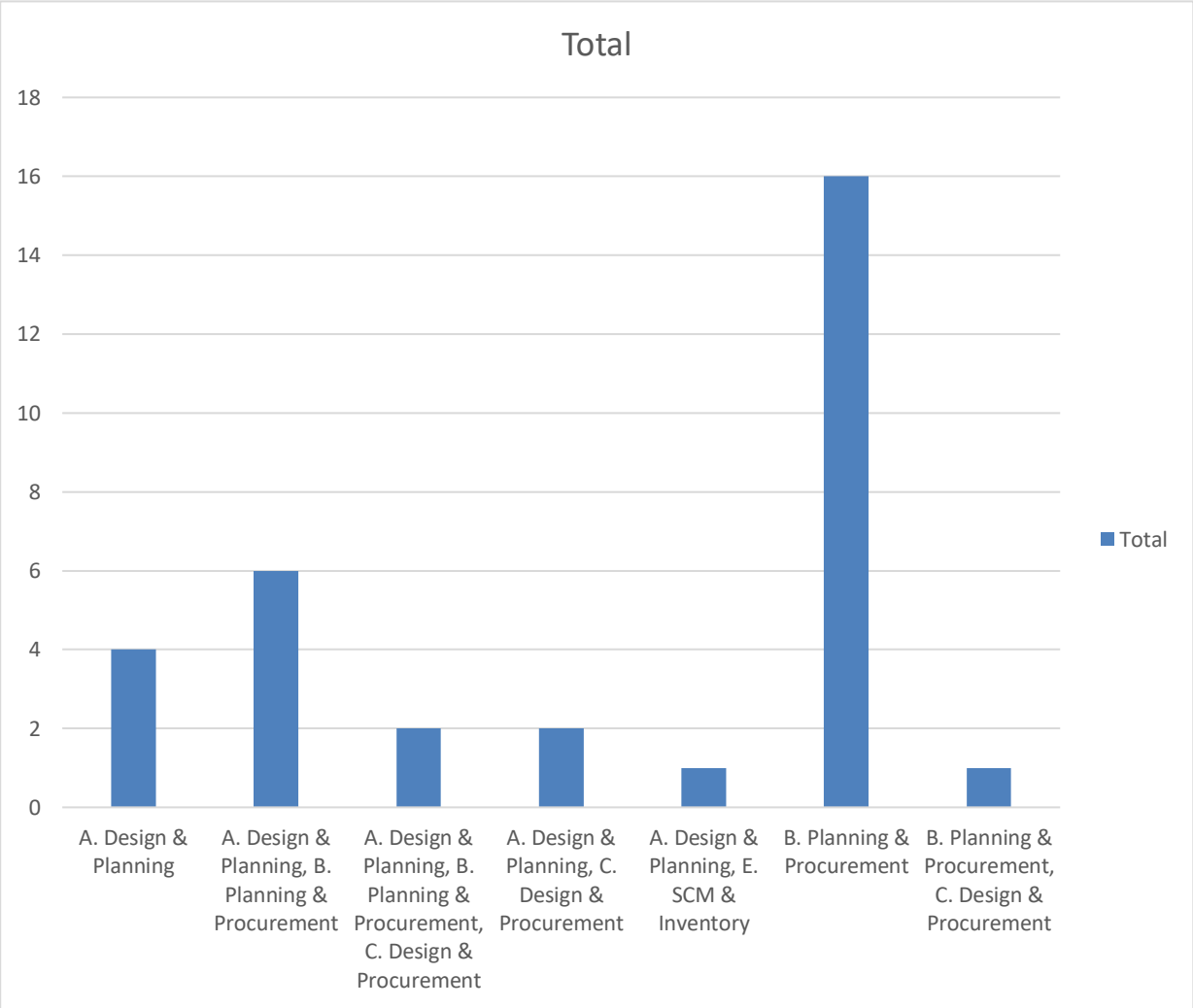
Interpretation:

Most respondents (71.9%) rely on Excel/manual tracking, either alone or with email threads, indicating a preference for manual methods. Only 3.1% use solely an ERP system, highlighting limited automation in the PR to PO tracking process and potential areas for digital improvement.

Table 4.6.10
Frequencies of 10. Which departments have the most communication challenges?

Row Labels	Counts	% of Total	Cumulative %
A. Design & Planning	4	12.5%	12.5%
A. Design & Planning, B. Planning & Procurement	6	18.8%	31.3%
A. Design & Planning, B. Planning & Procurement, C. Design & Procurement	2	6.3%	37.5%
A. Design & Planning, C. Design & Procurement	2	6.3%	43.8%
A. Design & Planning, E. SCM & Inventory	1	3.1%	46.9%
B. Planning & Procurement	16	50.0%	96.9%
B. Planning & Procurement, C. Design & Procurement	1	3.1%	100.0%

Graph 4.6.10
communication challenges



Interpretation:

Half of the respondents (50%) identified Planning & Procurement as the departments facing the most communication challenges. Additionally, 43.8% noted issues involving Design & Planning, either alone or with others, indicate that coordination among these core departments is a significant area for improvement.

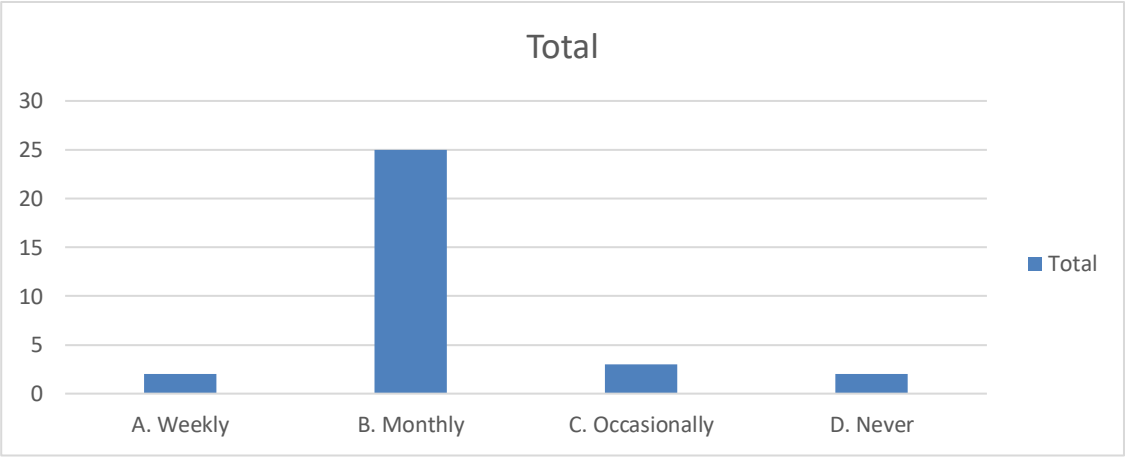
Table 4.6.11

Frequencies of 11. How often do inter-department meetings occur for PO planning?

Row Labels	Counts	% of Total	Cumulative %
A. Weekly	2	6.3%	6.3%
B. Monthly	25	78.1%	84.4%
C. Occasionally	3	9.4%	93.8%
D. Never	2	6.3%	100.0%

Graph 4.6.11

inter-department meetings



Interpretation

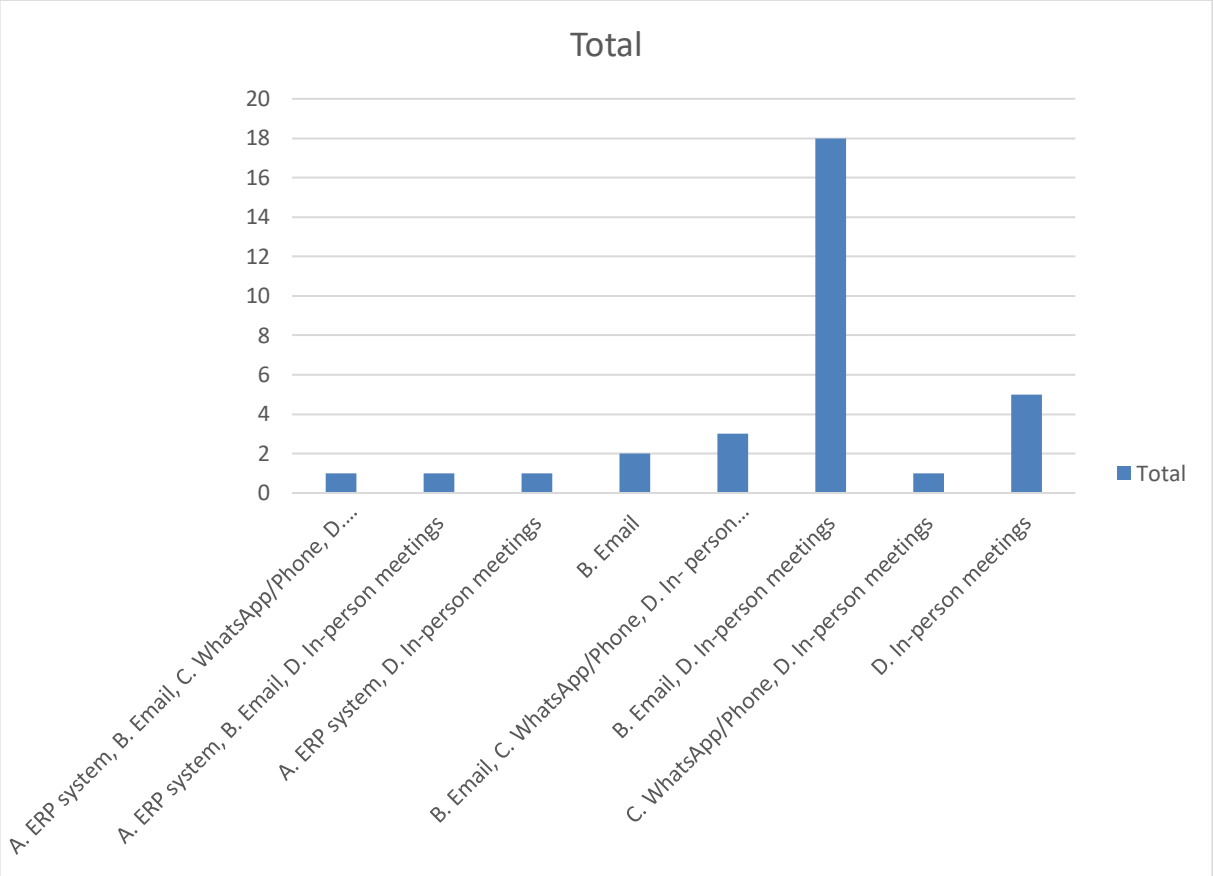
Most respondents (78.1%) reported that inter-department meetings for PO planning occur monthly, indicating a consistent but not frequent communication pattern. Only 6.3% hold weekly meetings, while 15.7% meet occasionally or never, suggesting room to enhance collaboration through more regular interactions.

Table 4.6.12
Frequencies of 12. What is the primary method of communication between departments

Row Labels	Counts	% of Total	Cumulative %
A. ERP system, B. Email, C. WhatsApp/Phone, D. In-person meetings	1	3.1%	3.1%
A. ERP system, B. Email, D. In-person meetings	1	3.1%	6.3%
A. ERP system, D. In-person meetings	1	3.1%	9.4%
B. Email	2	6.3%	15.6%
B. Email, C. WhatsApp/Phone, D. In-person meetings	3	9.4%	25.0%
B. Email, D. In-person meetings	18	56.3%	81.3%
C. WhatsApp/Phone, D. In-person meetings	1	3.1%	84.4%
D. In-person meetings	5	15.6%	100.0%

Graph 4.6.12

communication between departments



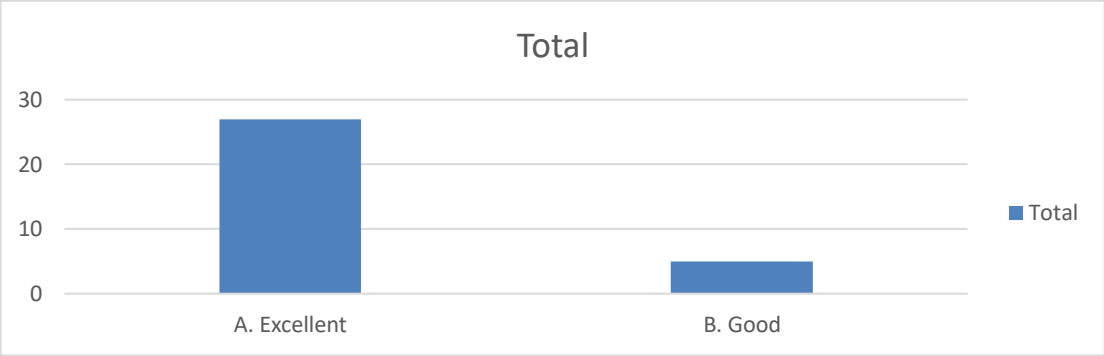
Interpretation:

The primary method of inter-department communication is a combination of **email and in-person meetings**, reported by 56.3% of respondents. While other methods like WhatsApp/Phone and ERP are used, they are far less common, indicating a strong reliance on traditional communication channels for coordination.

Table 4.6.13
Frequencies of 13. How would you rate your department's access to information from other departments?

Frequencies of 13. How would you rate your department's access to information from other departments?			
Row Labels	**Counts**	**% of Total**	**Cumulative %**
A. Excellent	**27**	**84.4%**	**84.4%**
B. Good	**5**	**15.6%**	**100.0%**

Graph 4.6.13
access to information



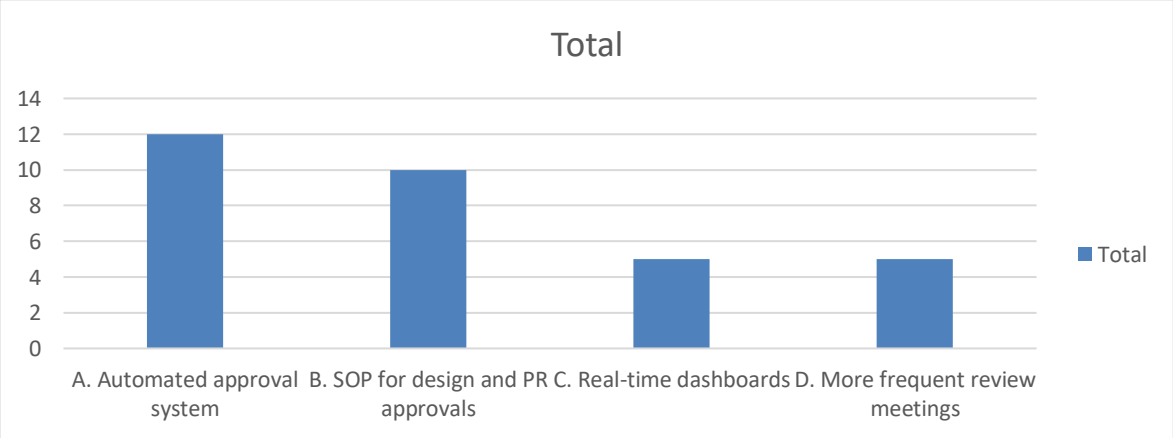
Interpretation:

An overwhelming majority (84.4%) rated their department's access to information from other departments as excellent, while the remaining 15.6% rated it as good. This indicates a generally strong level of transparency and inter-departmental information flow across the organization.

Table 4.6.1
Frequencies of 14. Which of the following would most improve the PR to PO process

Row Labels	Counts	% of Total	Cumulative %
A. Automated approval system	12	37.5%	37.5%
B. SOP for design and PR approvals	10	31.3%	68.8%
C. Real-time dashboards	5	15.6%	84.4%
D. More frequent review meetings	5	15.6%	100.0%

Graph 4.6.14
improve the PR to PO process



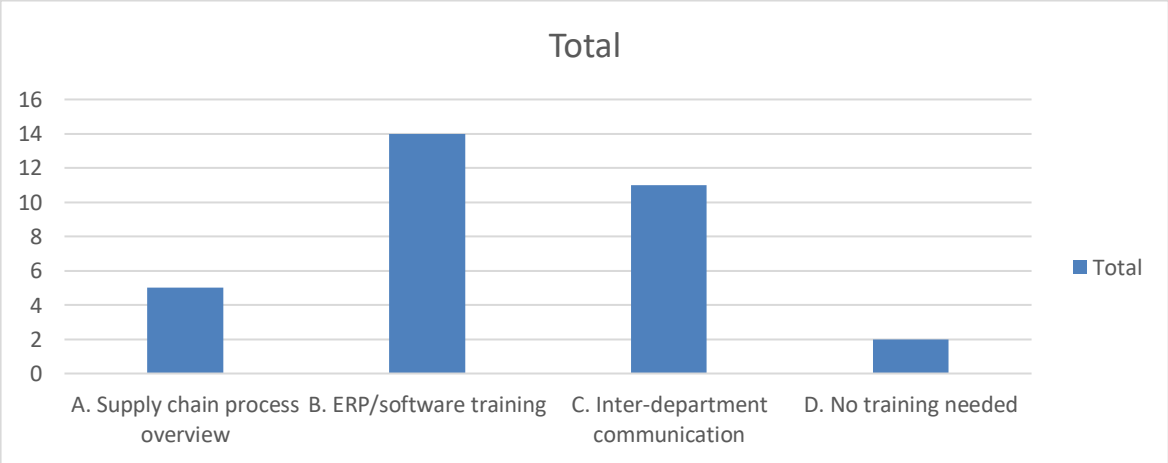
Interpretation:

The most suggested improvement for the PR to PO process is implementing an automated approval system (37.5%), followed by establishing SOPs for design and PR approvals (31.3%). Real-time dashboards and more frequent meetings were also noted (15.6% each), highlighting a desire for both process automation and clearer procedural structure.

Table 4.6.15
Frequencies of 15. What kind of training would help your department improve pre-order efficiency?

Row Labels	Counts	% of Total	Cumulative %
A. Supply chain process overview	5	15.6%	15.6%
B. ERP/software training	14	43.8%	59.4%
C. Inter-department communication	11	34.4%	93.8%
D. No training needed	2	6.3%	100.0%

Graph 4.6.15
Training Help



Interpretation:

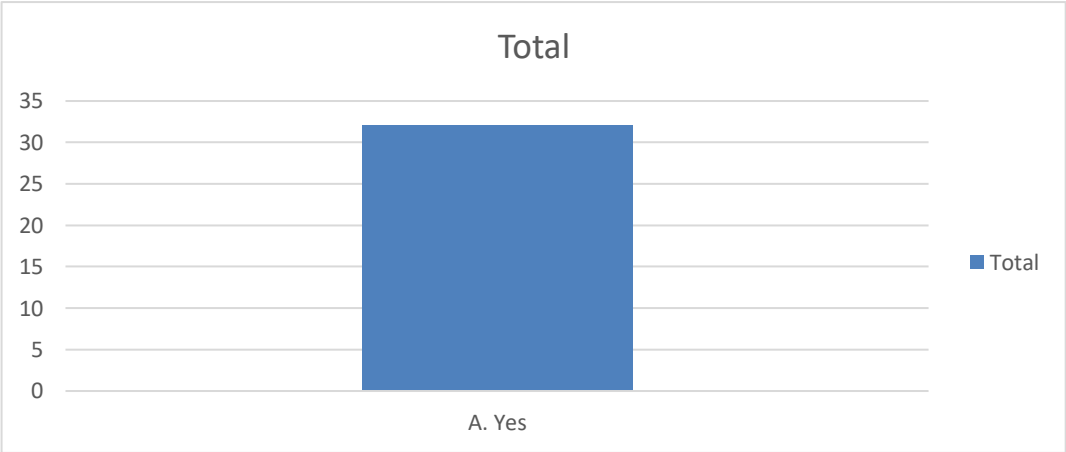
The most requested training to improve pre-order efficiency is ERP/software training (43.8%), followed by inter-department communication (34.4%). A smaller group (15.6%) sees value in a broader supply chain process overview, while only 6.3% believe no training is needed, indicating a strong demand for skill enhancement.

Table 4.6.16
Frequencies of 16. Do you think automating the PR to PO process will improve performance?

Row Labels	Counts	% of Total	Cumulative %
A. Yes	32	100.0%	100.0%

Grap 4.6.16

improve performance



Interpretation:

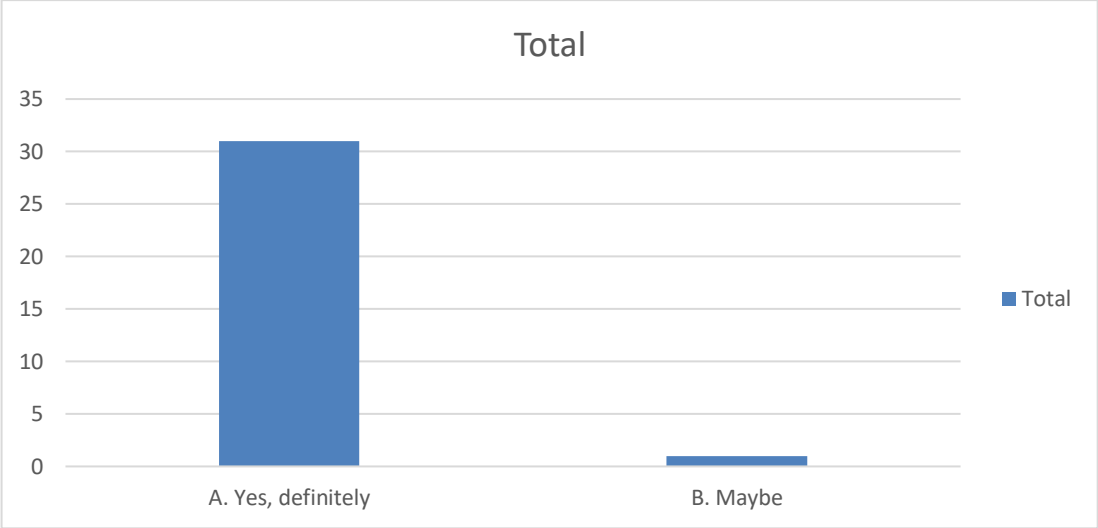
All respondents (100%) unanimously agreed that automating the PR to PO process would improve performance. This highlights a clear and collective recognition of the need for digital transformation to enhance efficiency and reduce manual bottlenecks.

Table 4.6.17
Frequencies of 17. Would a digital workflow reduce manual errors and approval delays?

Row Labels	Counts	% of Total	Cumulative %
A. Yes, definitely	31	96.9%	96.9%
B. Maybe	1	3.1%	100.0%

Graph 4.6.17

Reduce manual errors and approval delay



Interpretation:

An overwhelming majority (96.9%) believe that a digital workflow would reduce manual errors and approval delays, indicating strong confidence in digital solutions. Only 3.1% were uncertain, showing near-universal support for workflow automation.

Contingency Tables- (chi- Square)

Table 4.6.17
5. Which pre-requisite activity causes the most delay?

Designation	A. Forecasting issues	A. Forecasting issues, B. Delayed design finalization	B. Delayed design finalization	B. Delayed design finalization, Coinventor misalignment	C.Inventory misalignment	Total
A. Manager	0	0	1	0	1	2
B. Asst. Manager	0	1	3	0	0	4
C. Senior manager	0	0	4	0	0	4

D. Engineer	**0**	**1**	**8**	**1**	**0**	**10**
F. Junior Officer	**2**	**3**	**5**	**2**	**0**	**12**
Total	**2**	**5**	**21**	**3**	**1**	**32**

χ^2 Tests			
	Value	**df**	**p**
χ^2	**24.2**	**16**	**0.085**
N	**32**		

Interpretation:

Delayed design finalization is the leading cause of pre-order delays, reported by 65.6% of respondents, especially among Engineers and Junior Officers. Combined issues with forecasting and design also contribute notably. Forecasting alone and inventory misalignment were less frequently cited. This highlights the need to prioritize timely design finalization.

Table 4.6.18

7.What is the most common cause of delay in the PR to PO process?

Delegation	A. Manager	B. Asst. Manager	C. Senior manager	D. Engineer	F. Junior Officer	Total
A. Pending approvals	0	1	0	0	1	2
A. Pending approvals, B. Incomplete PR documentation	0	0	0	2	0	2
A. Pending approvals, B. Incomplete PR documentation, C. Design changes	0	0	0	0	1	1
A. Pending approvals, B. Incomplete PR documentation, C. Design changes, D. Lack	0	0	0	0	1	1

of coordination						
A. Pending approvals, C. Design changes	**0**	**0**	**2**	**3**	**1**	**6**
A. Pending approvals, C. Design changes, D. Lack of coordination	**0**	**0**	**0**	**0**	**1**	**1**
B. Incomplete PR documentation	**1**	**2**	**1**	**0**	**4**	**8**
B. Incomplete PR documentation, C. Design changes	**0**	**0**	**1**	**2**	**2**	**5**
B. Incomplete PR documentation,	**0**	**0**	**0**	**0**	**1**	**1**

D. Lack of coordination						
C. Design changes	**1**	**1**	**0**	**3**	**0**	**5**
Total	**2**	**4**	**4**	**10**	**12**	**32**

χ^2 Tests			
	Value	**df**	**p**
χ^2	**30.2**	**36**	**0.742**
N	**32**		

Interpretation:

The most common cause of delay in the PR to PO process is incomplete PR documentation, either alone or combined with other factors, affecting over 50% of respondents. This issue is particularly reported by Junior Officers and Engineers. Design changes and pending approvals are also frequent contributors. Managers and senior staff report fewer issues, indicating delays are more visible at operational levels. Improving PR completeness and design stability could significantly reduce delays.

4.7. TEST USED

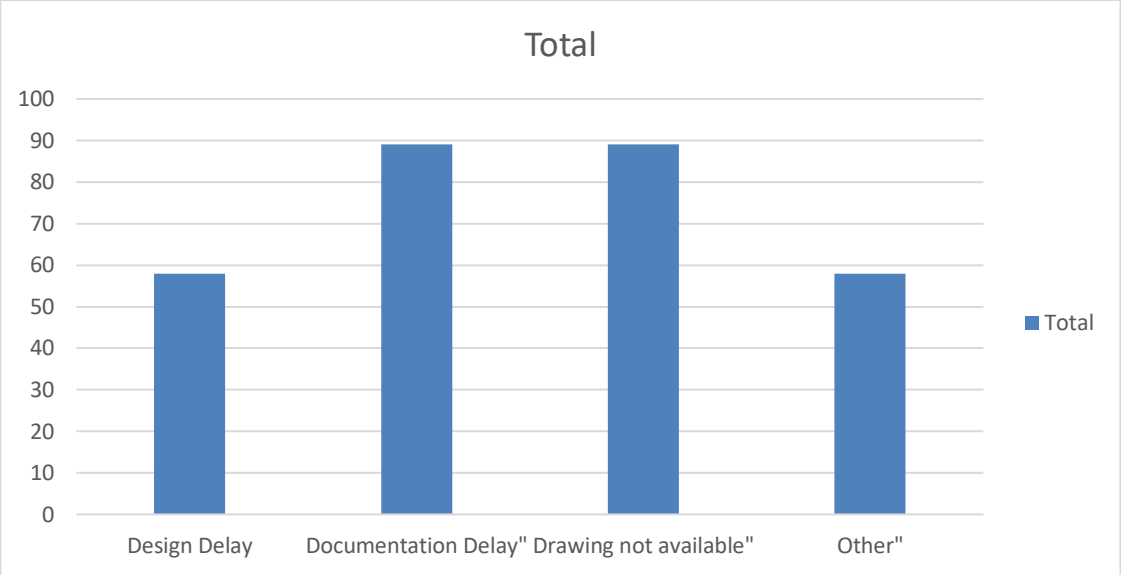
1. **Frequency/ Chi-square-**

Table 4.7.1

Frequency/ Chi-square-

Delay type	Counts	% of Total	Cumulative %
Documentation Delay"	86	29.3%	29.3%
Drawing not available"	82	27.9%	57.1%
Design Delay	55	18.7%	75.9%
Other"	58	19.7%	95.6%
Drawing not available"	7	2.4%	98.0%
Design Delay	3	1.0%	99.0%
Documentation Delay"	3	1.0%	100.0%

Graph 4.7.1
Frequency/ Chi-square



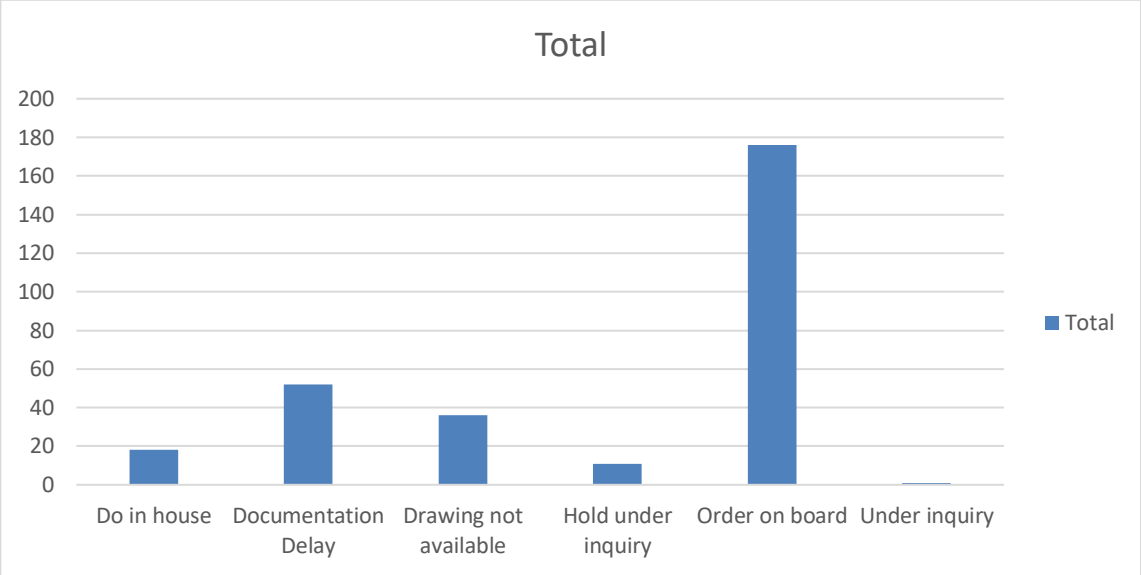
Interpretation:

The data shows that **Documentation Delay** (29.3%) and **Drawing Not Available** (27.9%) are the most frequent causes of delays, together accounting for over **57%** of total delays. **Design Delay** contributes **18.7%**, indicating it's also a significant factor. The remaining delays are relatively minor, with some entries appearing duplicated with lower frequencies, this may suggest issues in data classification or entry that need attention.

Table 4.7.2
Frequency

Frequencies-	Counts	% of Total	Cumulative %
Do in house	18	6.1%	6.1%
Documentation Delay	52	17.7%	23.8%
Drawing not available	36	12.2%	36.1%
Hold under inquiry	11	3.7%	39.8%
Order on board	176	59.9%	99.7%
Under inquiry	1	0.3%	100.0%

Graph 4.7.2
Frequency



Interpretation:

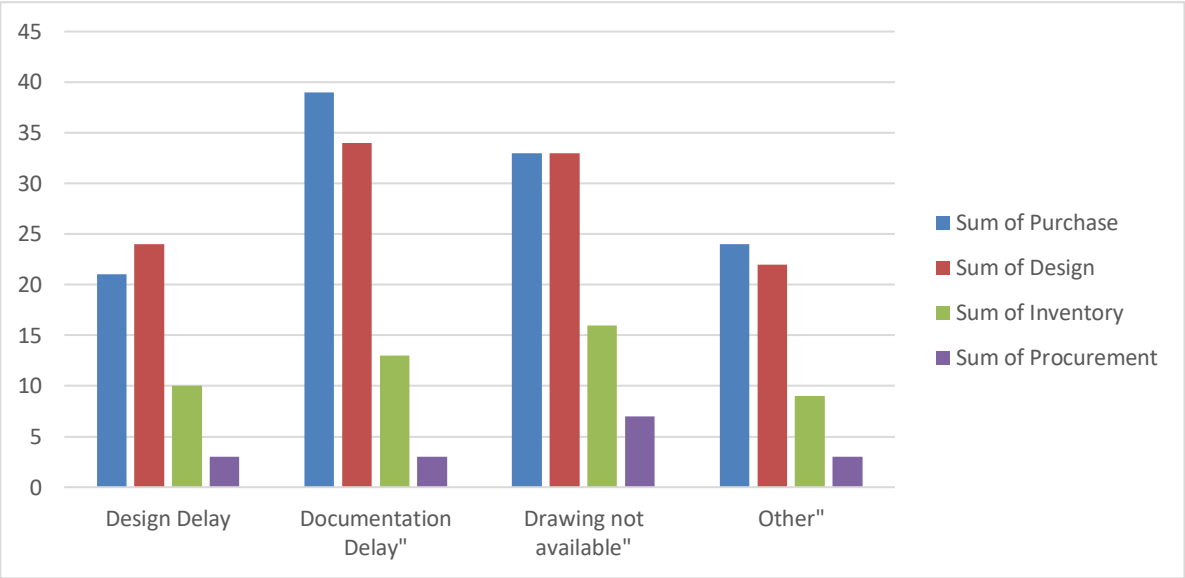
The frequency data indicates that most occurrences (**59.9%**) are concentrated in a single category, suggesting a strong dominance or common trend. The next most frequent category accounts for **17.7%**, while the remaining categories contribute minimally. This skewed distribution implies that one issue or behaviour overwhelmingly influences the total and should be the primary focus for further analysis or corrective action.

Table 4.7.3

Chi-square

Delay type	Purchase	Procurement	Design	Inventory	Total
Documentation Delay"	39	0	34	13	86
Drawing not available"	33	0	33	16	82
Design Delay	21	0	24	10	55
Other"	24	3	22	9	58
Drawing not available"	0	7	0	0	7
Design Delay	0	3	0	0	3
Documentation Delay"	0	3	0	0	3
Total	117	16	113	48	294

Graph 4.7.2
Chi-square



Interpretation:

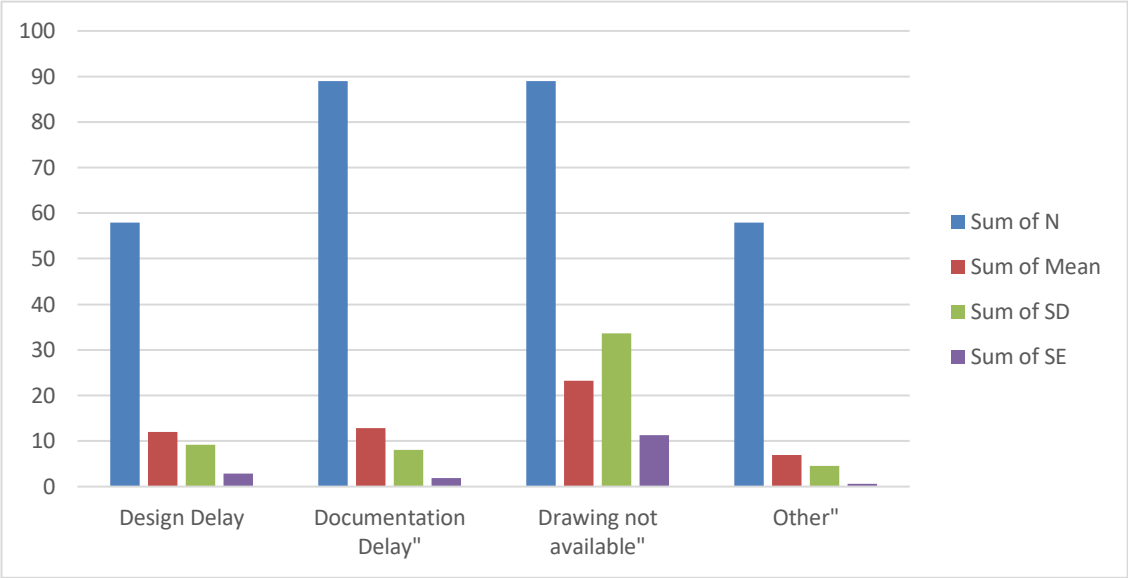
The contingency table reveals that Purchase and Design departments are most affected by delays, with high counts across major delay types. Specifically, Documentation Delay and Drawing Not Available are prominent in both Purchase (39, 33) and Design (34, 33). Procurement shows minimal involvement, mostly with less delay types. Inventory faces moderate delays, mainly due to Documentation and Drawing availability issues. The data highlights that Purchase and Design departments should be prioritized for process improvements to reduce overall delays.

1. One-Way ANOVA

Table 4.7.6
One-Way ANOVA

Delay type	**N**	**Mean**	**SD**	**SE**
Documentation Delay"	**86**	**7.12**	**5.79**	**0.624**
Drawing not available"	**82**	**6.99**	**5.37**	**0.593**
Design Delay	**55**	**7.21**	**5.69**	**0.767**
Other"	**58**	**6.88**	**4.61**	**0.606**
Drawing not available"	**7**	**16.21**	**28.20**	**10.660**
Design Delay	**3**	**4.74**	**3.53**	**2.037**
Documentation Delay"	**3**	**5.77**	**2.25**	**1.298**

Graph 4.7.4
One-Way ANOVA



Interpretation:

The group descriptives show that the **average ageing (delay duration)** is consistent across the main delay types—around **7 days** for **Documentation Delay**, **drawing not available**, and **Design Delay**, with standard deviations indicating moderate variability. However, the second entry of **Drawing not available** (N=7) has a **much higher mean (16.21)** and **very high variability (SD=28.20)**, suggesting extreme outliers or exceptional cases. The smaller groups (N=3) also show less reliability due to higher standard errors. This points to the need for data cleaning or separation of duplicate categories for clearer analysis.

4.8. FINDING

A. **Pre-Requisite Activities and Delays**

a. The most significant pre-order delay is caused by delayed design finalization, as reported by 65.6% of respondents, especially among Engineers and Junior Officers.

b. When it comes to delays in the PR to PO process, the top issue is incomplete PR documentation, affecting more than half the respondents.

c. Other contributing factors include frequent design changes and pending approvals, with operational-level staff being most impacted by these challenges.

B. Delay Frequency and Impact

a. Documentation Delay (29.3%) and Drawing Not Available (27.9%) together make up over 57% of the total delays, indicating a clear pattern.

b. Design Delay adds another 18.7%, showing it's a consistent process gap.

c. Aging analysis reveals that most delays average around 7 days, but a few outliers like one case with over 16 days and high variability indicate possible data entry issues or exceptional cases that need review.

D. Departmental Impact

a. The Purchase and Design departments are the most affected by delays, especially with documentation and drawings.

b. While Procurement and Inventory departments face fewer issues, they still experience moderate delays, primarily related to documentation.

E. Process Efficiency and Automation Needs

a. There is a clear consensus (100%) among respondents that automating the PR to PO process would significantly improve performance.

b. Nearly all (96.9%) agree that a digital workflow would help reduce manual errors and approval delays.

c. Currently, most tracking is done manually 71.9% rely on Excel or email pointing to an urgent need for better automation tools.

F. Communication and Coordination

a. Planning & Procurement is seen as the area with the most communication issues, followed closely by Design & Planning.

b. Most departments rely on email and face-to-face meetings (56.3%) for communication, with minimal use of ERP systems.

c. Inter-department meetings are held monthly in 78.1% of cases, while only 6.3% meet weekly, suggesting that more frequent coordination could improve efficiency.

G. Training and Improvement Opportunities

a. The top training need identified is ERP/software usage (43.8%), followed by inter-department communication skills (34.4%).

b. The most recommended improvements include setting up an automated approval system (37.5%) and creating SOPs for design and PR approvals (31.3%).

H. Information Access and Forecasting

a. A strong majority (84.4%) rated their access to information from other departments as excellent, showing good internal visibility.

b. Demand forecasting is regularly and accurately done in 65.6% of departments, but 31.3% still follow it occasionally indicating an opportunity for more consistent planning.

H. Inventory Planning and Processing Time

a. All respondents confirmed that inventory planning is completed before PR creation, demonstrating strong process discipline.

b. The PR to PO processing time is typically between 4 to 7 days (84.4%), reflecting an efficient cycle with minimal delays overall.

5. STUDENT LEARNING OUTCOMES-

5.1.TECHNICAL / FUNCTIONAL SKILLS GAINED-

During my OJT, I gained hands-on experience in understanding the operational workflow of the supply chain particularly the process from Purchase Requisition (PR) to Purchase Order (PO). I was able to observe and document how procurement requests originate, how approvals are structured, and how purchase orders are finalized. This helped me understand how inventory planning, material requirement planning (MRP), and vendor coordination align with the organization's procurement policies. As an MBA student, this exposure translated academic concepts like process mapping, procurement cycles, and lead time management into real industrial practices, strengthening my technical competence in operations and supply chain domains.

5.2.PROFESSIONAL & INTERPERSONAL SKILLS DEVELOPED

5.2.1. SOFT SKILLS

The OJT experience significantly contributed to improving my professional behaviour and interpersonal communication. I learned how to:

A. Draft structured and professional emails for internal communication.

B. Participate in meetings effectively, listening actively and contributing when required.

C. Follow up respectfully with team members and supervisors, reinforcing the importance of accountability in workplace communication.

5.2.2. TEAMWORK AND COLLABORATION-

Working across departments such as design, planning, and procurement gave me the opportunity to observe cross-functional teamwork. I learned how collaboration, timely communication, and alignment on goals are essential for operational efficiency, especially in a manufacturing setup

5.2.3. BUSINESS PROCESSES / PRACTICES UNDERSTOOD

A. To clear understanding of how interdepartmental coordination directly influences procurement speed, decision-making, and production timelines.

B. To observe the **real-time challenges** businesses, face such as delays in approvals, unclear responsibilities, and data discrepancies which can disrupt supply chain flow.

C. This practical learning helped me appreciate how theoretical models like supply chain frameworks, lean processes, and workflow optimization apply in the real world.

5.2.4. UNDERSTANDING OF WORKPLACE CULTURE-

A. **Hierarchy and Approvals**: I learned how formal approval structures ensure accountability and prevent errors. Every step in procurement, especially in high-value items—required multiple levels of review.

B. **Responsibility and Accountability**: Documentation, traceability, and maintaining proper records were considered vital to business integrity and compliance.

C. **Safety and Discipline**: I experienced the importance of punctuality, safety protocols, and professional conduct of which are strictly enforced in a manufacturing environment.

A. **Team Dynamics**: Informal team collaboration, respect for roles and experience, and professional bonding were noticeable elements that made teamwork smoother and more efficient.

5.2.5. AREAS FOR IMPROVEMENT

A. **Data Analysis Speed and Accuracy**: I need to improve my proficiency in tools like Excel and Power BI to analyse procurement data more effectively.

B. **Recognition of Efforts**: Understanding that continuous improvement and visibility of contribution are necessary for professional growth.

C. **Work Management**: Managing multiple tasks simultaneously under time constraints was a challenge that I plan to overcome by improving my organizational and time management skills.

5.3.FUTURE OPPORTUNITIES

This OJT experience has played a crucial role in shaping my long-term career aspirations. As an MBA student aiming to specialize in operations or business analytics, I discovered that the role of a **Business Analyst** aligns closely with my interests and strengths. Analysing delays in the PR-to-PO process gave me real insight into how data-backed decisions improve efficiency. I realized the importance of converting operational data into actionable insights, recommending process improvements, and facilitating communication between departments—core functions of a Business Analyst.

This internship has encouraged me to pursue a career path where I can work on process improvement, performance measurement, and cross-functional analysis. It has also motivated me to build on technical tools (Excel, ERP systems, process modelling) and soft skills (problem-solving, communication, coordination). Most importantly, it confirmed that I want to contribute meaningfully to business growth by bridging the gap between operations and strategy.

6. ENVIRONMENTAL, SAFETY, AND ETHICAL CONSIDERATIONS-

6.1. SAFETY

A. Wearing Personal Protective Equipment (PPE) such as safety shoes, helmets, and safety glasses within shop floors.
B. Attending a safety induction session on workplace hazards, emergency exits, and equipment handling.
C. Following Standard Operating Procedures (SOPs) for restricted areas and observing safety signage. ((Cesare Bonetti, n.d.))
D. Participating in fire safety drills and understanding emergency evacuation protocols.

6.2. ENVIRONMENTAL

6.2.1. RESPONSIBILITIES

Management (the supervisor and / or manager) is responsible for providing and maintaining:

A. A safe working environment
B. Safe systems of work and Facilities for the welfare of Employees
C. Plant and substances in safe condition
D. Information, instruction, training and supervision that is reasonably necessary to ensure that each employee is safe from injury and risks to health
E. A commitment to consult and co-operate with employees in all matters relating to Environment health and safety in the workplace
F. A commitment to continual improvement of our performance through effective safety management.
G. Any information, instruction, training and supervision needed to make sure that all employees are safe from injury and risks to their health. (Cesare Bonetti, 2025)

6.2.2. EMPLOYEES ARE RESPONSIBLE FOR

A. Comply with safe work practices, with the intent of avoiding injury to themselves and others and damage to plant and equipment.
B. Take reasonable care of the health and safety of themselves and others

C. Wear personal protective equipment and clothing where necessary

D. Complying with any reasonable directions (such as safe work procedures, and wearing personal protective equipment) given by management for health and safety

E. No misuse or interfere with anything provided for health and safety

F. Report all accidents and incidents on the job immediately, no matter how trivia

G. Report all known or observed hazards to their supervisor or manager.

6.2.3. ETHICALS

A. **Transparency in Process**: All procurement activities follow formal documentation, audit trails, and approval hierarchies.

B. **Data Confidentiality**: Interns and employees are given access to only relevant, non-sensitive information. Confidential data is protected and not shared outside authorized teams.

C. **Equal Treatment**: Professional behaviour, respect for all employees regardless of role, and a harassment-free workplace were strictly maintained.

D. **Integrity in Operations**: Decision-making was based on factual data, technical evaluation, and approved vendor processes avoiding favouritism or bias. ((Cesare Bonetti, 2025))

6.2.4. COST-EFFECTIVENESS

A. **Vendor evaluation based on value**, not just price, to ensure long-term performance and warranty benefits.

B. **Avoidance of rework and delays** by ensuring proper design approval and accurate requirement planning in the pre-order phase.

C. **Streamlined processes** that reduce idle time, avoid overstocking, and control procurement costs.

6.2.5. PROFESSIONAL STANDARDS (National/International)

A. ISO 9001: For quality management systems.

B. ASME, DIN, and EN standards: For valve and pressure equipment design,

manufacturing, and testing.

C. **Indian Standard Procurement Norms**: For fair vendor selection and transparent internal processes.

D. **Internal Audits and SOPs**: Ensuring consistent compliance with operational and safety benchmarks. (Cesare Bonetti,2025)

7. CONTRIBUTION TO THE HOST ORGANIZATION

7.1.PROJECTS/TASKS COMPLETED THAT ADDED VALUE

A. **Department-Wise Survey**: Collected and analysed responses from 32 employees across departments (Design, PPC, Inventory, and Purchase) to identify inefficiencies in the PR to PO cycle.

B. **Data Compilation and Visualization**: Created structured tables covering variables such as gender, department, designation, experience, and various process-specific metrics.

C. **Process Gap Identification**: Mapped the existing PR-to-PO process and highlighted bottlenecks, especially in areas such as delayed design finalization, incomplete PR documentation, and manual tracking.

D. **Root Cause Analysis**: Applied interpretive techniques to pinpoint that *81.25%* of delays were due to **design finalization issues**, and *37.5%* of respondents identified **incomplete PR documentation** as a major cause of delay.

E. **Digital Transformation Proposal**: Based on findings, proposed an automation strategy to reduce manual intervention, streamline approvals, and ensure real- time tracking.

7.2. IMPROVEMENTS/SUGGESTIONS IMPLEMENTED

A. **Proposal for an Automated PR Approval Workflow**: Recommended automation of PR approval steps through ERP **(Enterprise Resource Planning)** integration. This was backed by 100% of respondents who agreed automation would improve performance.

B. **Training Recommendations**: Suggested ERP and communication training, supported by 43.75% and 34.38% of participants respectively, to improve efficiency in pre-order processing.

C. **Standard Operating Procedures (SOPs)**: Advised creation of SOPs for design and PR approvals to ensure consistent documentation quality and reduce rework.

REFERENCE

A. **Effect of Pre-Contracting Procurement Practices on Supply Chain Performance of Road Authorities in Kakamega County, Kenya**Okinyi N, Kwendo E, Kiongera F*African Journal of Empirical Research 4 807-814*

B. **Analytics for Supply Chain Optimisation**Uttarwar H

C. **Integrated Supply Chain Performance Measurement Model for the Manufacturing Industry**Saleheen F, Habib M*Journal of Service Science and Management (2022) 15(02) 55-70*

D. **EPiC Series in Computing Detection of Bottleneck in Manufacturing Supply Chain using Specific KPI**Kawabata Y, Hosokawa Y, Fujita K*(2022)*

E. **Material Ordering and Transferring Plan Development Based on 0-1 Planning Model***Academic Journal of Computing & Information Science (2022) 5(1)*

F. **The Optimal Supply Decision Based on Dynamic Multiobjective Optimization and Prediction** He K, Cheng H, Xie C*Journal of Function Spaces (2022) 2022*

G. **Supplier selection and procurement in SMEs: Insights from the literature on key criteria and purchasing strategies**Ferreira A, Silva Â*Engineering Management in Production and Services (2022) 14(4) 47-60*

H. **Automation Work Order And Master Production Schedule In Enterprise Resources Planning For Pt Sfp**Reka N*IT FOR SOCIETY 08(02)*

I. **Journal of Transport and Supply Chain Management**Klein Y, Lubinga S, Masiya T*(2024)*

Website

Cesare Bonetti. (n.d.). *High performanc0e valves & liquid level gauges*. Retrieved July 28, 2025, from https://c-bonetti.com/

APPENDICES

QUESTIONS

Respondent Information-

1.Name

_____

2.Gender

A. Male

B. Female

3.Department

C. Design

D. Planning

E. Procurement

F. Inventory

G. Purchase

4. Designation

H. Manager

I. Asst. Manager

J. Executive

K. Jr. Executive

L. Other

5. Years of experience in supply chain-related functions.

M. Less than 1 year

N. 1–3 years

O. 3–5 years

P. More than 5 years

Objective 1- To identify and analyse the key pre-requisite activities (such as demand forecasting, inventory planning, and design finalization) that affect the timely release of purchase orders.

Hypothesis 1

A. H_0- Demand forecasting, inventory planning, and design finalization do not significantly affect the timely release of purchase orders.

B. H_1- Demand forecasting, inventory planning, and design finalization significantly affect the timely release of purchase orders.

Q1. What is the status of demand forecasting in your department?

Objective: 1 | Hypothesis: 1

A. Regular and accurate

B. Done occasionally

C. Not done at all

D. Not applicable to my department

Q2. When is inventory planning usually done?

Objective: 1 | Hypothesis: 1

A. Before PR is created

B. After PR

D. Not planned systematically

Q3. When is design finalization usually completed?

Objective: 1 | Hypothesis: 1

A. Before PR

B. Between PR and PO

C. After PO

D. It varies

Q4. Which pre-requisite activity causes the most delay?

Objective: 1 | Hypothesis: 1

A. Forecasting issues

B. Delayed design finalization

C. Inventory misalignment

D. All the above

Q5. Wat is the average processing time from PR creation to PO release?

Objective: 1 | Hypothesis: 1

A. Same day

B. 1-3 days

C. 4-7 days

D. 1-2 weeks

E. More than 2 weeks

Objective 2- To examine the common issues and delays in the pre-ordering phase, particularly in the transition from Purchase Requisition (PR) to Purchase Order (PO).

Hypothesis 2

A. H_0- Delays in the transition from PR to PO do not impact the overall supply chain efficiency.

B. H_1- Delays in the transition from PR to PO negatively impact the overall supply chain efficiency.

Q6. How often do PRs get delayed in approval?

Objective: 2 | Hypothesis: 2

A. Very Frequently

B. Sometimes

C. Rarely

D. Never

Q7. What is the most common cause of delay in the PR to PO process?

Objective: 2 | Hypothesis: 2

A. Pending approvals

B. Incomplete PR documentation

C. Design changes

D. Lack of coordination

Q8. Who typically initiates the follow-up for delayed approvals?

Objective: 2 | Hypothesis: 2

A. Design department

B. Planning department

C. Procurement team

D. No one follows up regularly

Q9. How is the PR to PO tracking done?

Objective: 2 | Hypothesis: 2

A. Through ERP system

B. Excel/manual tracking

C. Email threads

D. No defined tracking method

Objective 3- To evaluate the level of coordination and communication between departments such as design, planning, and procurement during the early stages of the supply chain process.

Hypothesis 3

A. H_0- Coordination between departments has no effect on the speed of the pre- ordering process.
B. H_1- Better coordination improves the speed of the pre-ordering process.

Q10. Which departments have the most communication challenges?

Objective: 3 | Hypothesis: 3

A. Design & Planning

B. Planning & Procurement

C. Design & Procurement

D. All equally

E. No major issues

Q11. How often do inter-department meetings occur for PO planning?

Objective: 3 | Hypothesis: 3

A. Weekly

B. Monthly

C. Occasionally

D. Never

Q12. What is the primary method of communication between departments?

Objective: 3 | Hypothesis: 3

A. ERP system

B. Email

C. WhatsApp/Phone

D. In-person meetings

Q13. How would you rate your department's access to information from other departments?

Objective: 3 | Hypothesis: 3

A. Excellent

B. Good

C. Average

D. Poor

E. Very Poor

Objective 4-To suggest practical process improvements for reducing delays, improving inter-departmental collaboration, and ensuring smoother execution of procurement operations.

Hypothesis 4

A. H_0- Improving the pre-ordering process does not lead to better on-time delivery of products.

B. H_1- Improving the pre-ordering process leads to better on-time delivery of products.

Q14. Which of the following would most improve the PR to PO process?

Objective: 4 | Hypothesis: 4

A. Automated approval system

B. SOP for design and PR approvals

C. Real-time dashboards

D. More frequent review meetings

Q15. What kind of training would help your department improve pre-order efficiency?

Objective: 4 | Hypothesis: 4

A. Supply chain process overview

B. ERP/software training

C. Inter-department communication

D. No training needed

Q16. Do you think automating the PR to PO process will improve performance?

Objective: 4 | Hypothesis: 4

A. Yes

B. No

C. Maybe

D. Not sure

Q17. Would a digital workflow reduce manual errors and approval delays?

Objective: 4 | Hypothesis: 4

A. Yes, definitely

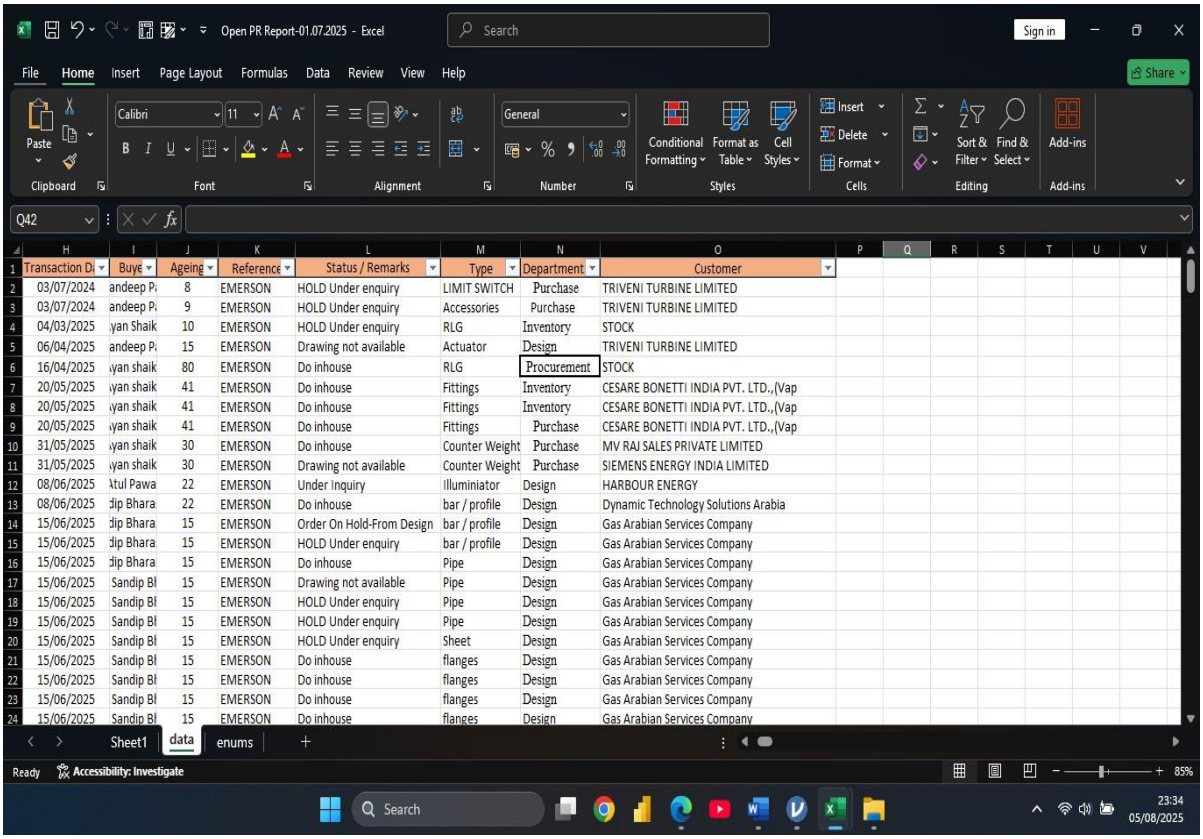
B. Maybe

C. No impact

D. Don’t know

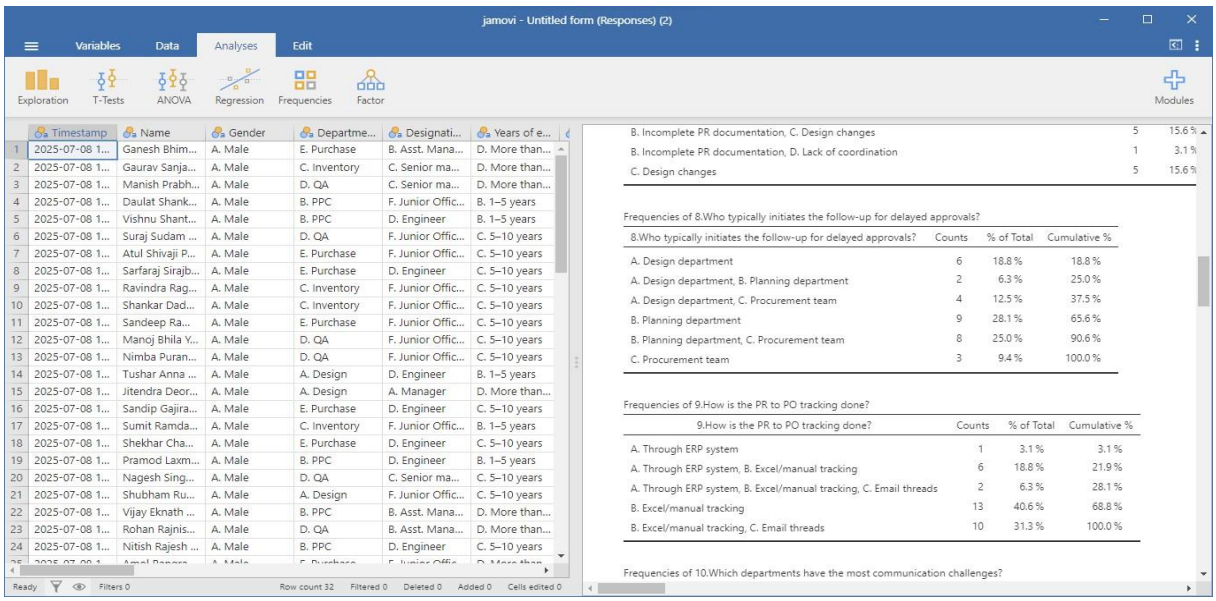
Secondary Data-

fig 1
Company Data in Excel



Transaction Da	Buye	Ageing	Reference	Status / Remarks	Type	Department	Customer
03/07/2024	andeep Pa	8	EMERSON	HOLD Under enquiry	LIMIT SWITCH	Purchase	TRIVENI TURBINE LIMITED
03/07/2024	andeep Pa	9	EMERSON	HOLD Under enquiry	Accessories	Purchase	TRIVENI TURBINE LIMITED
04/03/2025	yan Shaik	10	EMERSON	HOLD Under enquiry	RLG	Inventory	STOCK
06/04/2025	andeep Pa	15	EMERSON	Drawing not available	Actuator	Design	TRIVENI TURBINE LIMITED
16/04/2025	yan shaik	80	EMERSON	Do inhouse	RLG	Procurement	STOCK
20/05/2025	yan shaik	41	EMERSON	Do inhouse	Fittings	Inventory	CESARE BONETTI INDIA PVT. LTD.,(Vap
20/05/2025	yan shaik	41	EMERSON	Do inhouse	Fittings	Inventory	CESARE BONETTI INDIA PVT. LTD.,(Vap
20/05/2025	yan shaik	41	EMERSON	Do inhouse	Fittings	Purchase	CESARE BONETTI INDIA PVT. LTD.,(Vap
31/05/2025	yan shaik	30	EMERSON	Do inhouse	Counter Weight	Purchase	MV RAJ SALES PRIVATE LIMITED
31/05/2025	yan shaik	30	EMERSON	Drawing not available	Counter Weight	Purchase	SIEMENS ENERGY INDIA LIMITED
08/06/2025	tul Pawa	22	EMERSON	Under Inquiry	Illuminiator	Design	HARBOUR ENERGY
08/06/2025	dip Bhara	22	EMERSON	Do inhouse	bar / profile	Design	Dynamic Technology Solutions Arabia
15/06/2025	dip Bhara	15	EMERSON	Order On Hold-From Design	bar / profile	Design	Gas Arabian Services Company
15/06/2025	dip Bhara	15	EMERSON	HOLD Under enquiry	bar / profile	Design	Gas Arabian Services Company
15/06/2025	dip Bhara	15	EMERSON	Do inhouse	Pipe	Design	Gas Arabian Services Company
15/06/2025	Sandip Bh	15	EMERSON	Drawing not available	Pipe	Design	Gas Arabian Services Company
15/06/2025	Sandip Bh	15	EMERSON	HOLD Under enquiry	Pipe	Design	Gas Arabian Services Company
15/06/2025	Sandip Bh	15	EMERSON	HOLD Under enquiry	Pipe	Design	Gas Arabian Services Company
15/06/2025	Sandip Bh	15	EMERSON	HOLD Under enquiry	Sheet	Design	Gas Arabian Services Company
15/06/2025	Sandip Bh	15	EMERSON	Do inhouse	flanges	Design	Gas Arabian Services Company
15/06/2025	Sandip Bh	15	EMERSON	Do inhouse	flanges	Design	Gas Arabian Services Company
15/06/2025	Sandip Bh	15	EMERSON	Do inhouse	flanges	Design	Gas Arabian Services Company
15/06/2025	Sandip Bh	15	EMERSON	Do inhouse	flanges	Design	Gas Arabian Services Company

fig 2
Use Jamovi for Test Run



Timestamp	Name	Gender	Departme...	Designati...	Years of e...
2025-07-08 1...	Ganesh Bhim...	A. Male	E. Purchase	B. Asst. Mana...	D. More than...
2025-07-08 1...	Gaurav Sanja...	A. Male	C. Inventory	C. Senior ma...	D. More than...
2025-07-08 1...	Manish Prabh...	A. Male	D. QA	C. Senior ma...	D. More than...
2025-07-08 1...	Daulat Shank...	A. Male	B. PPC	F. Junior Offic...	B. 1–5 years
2025-07-08 1...	Vishnu Shant...	A. Male	B. PPC	D. Engineer	B. 1–5 years
2025-07-08 1...	Suraj Sudam ...	A. Male	D. QA	F. Junior Offic...	C. 5–10 years
2025-07-08 1...	Atul Shivaji P...	A. Male	E. Purchase	F. Junior Offic...	C. 5–10 years
2025-07-08 1...	Sarfaraj Sirajb...	A. Male	E. Purchase	D. Engineer	C. 5–10 years
2025-07-08 1...	Ravindra Rag...	A. Male	C. Inventory	F. Junior Offic...	C. 5–10 years
2025-07-08 1...	Shankar Dad...	A. Male	C. Inventory	F. Junior Offic...	C. 5–10 years
2025-07-08 1...	Sandeep Ra...	A. Male	E. Purchase	F. Junior Offic...	C. 5–10 years
2025-07-08 1...	Manoj Bhila Y...	A. Male	D. QA	F. Junior Offic...	C. 5–10 years
2025-07-08 1...	Nimba Puran...	A. Male	D. QA	F. Junior Offic...	C. 5–10 years
2025-07-08 1...	Tushar Anna ...	A. Male	A. Design	D. Engineer	B. 1–5 years
2025-07-08 1...	Jitendra Deor...	A. Male	A. Design	A. Manager	D. More than...
2025-07-08 1...	Sandip Gajira...	A. Male	E. Purchase	D. Engineer	C. 5–10 years
2025-07-08 1...	Sumit Ramda...	A. Male	C. Inventory	F. Junior Offic...	B. 1–5 years
2025-07-08 1...	Shekhar Cha...	A. Male	E. Purchase	D. Engineer	C. 5–10 years
2025-07-08 1...	Pramod Laxm...	A. Male	B. PPC	D. Engineer	B. 1–5 years
2025-07-08 1...	Nagesh Sing...	A. Male	D. QA	C. Senior ma...	C. 5–10 years
2025-07-08 1...	Shubham Ru...	A. Male	A. Design	F. Junior Offic...	C. 5–10 years
2025-07-08 1...	Vijay Eknath ...	A. Male	B. PPC	B. Asst. Mana...	D. More than...
2025-07-08 1...	Rohan Rajnis...	A. Male	D. QA	B. Asst. Mana...	D. More than...
2025-07-08 1...	Nitish Rajesh ...	A. Male	B. PPC	D. Engineer	C. 5–10 years

B. Incomplete PR documentation, C. Design changes	5	15.6 %
B. Incomplete PR documentation, D. Lack of coordination	1	3.1 %
C. Design changes	5	15.6 %

Frequencies of 8.Who typically initiates the follow-up for delayed approvals?

8.Who typically initiates the follow-up for delayed approvals?	Counts	% of Total	Cumulative %
A. Design department	6	18.8 %	18.8 %
A. Design department, B. Planning department	2	6.3 %	25.0 %
A. Design department, C. Procurement team	4	12.5 %	37.5 %
B. Planning department	9	28.1 %	65.6 %
B. Planning department, C. Procurement team	8	25.0 %	90.6 %
C. Procurement team	3	9.4 %	100.0 %

Frequencies of 9.How is the PR to PO tracking done?

9.How is the PR to PO tracking done?	Counts	% of Total	Cumulative %
A. Through ERP system	1	3.1 %	3.1 %
A. Through ERP system, B. Excel/manual tracking	6	18.8 %	21.9 %
A. Through ERP system, B. Excel/manual tracking, C. Email threads	2	6.3 %	28.1 %
B. Excel/manual tracking	13	40.6 %	68.8 %
B. Excel/manual tracking, C. Email threads	10	31.3 %	100.0 %

Frequencies of 10.Which departments have the most communication challenges?